

Interactive Multiscale Modeling to Bridge Atomic Properties and Electrochemical Performance in Li-CO₂ Battery Design

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Abstract

Li-CO₂ batteries are promising energy storage systems due to their high theoretical energy density and CO₂ fixation capability, relying on reversible Li₂CO₃/C formation during discharge/charge cycles. We present a multiscale modeling framework integrating Density Functional Theory (DFT), Ab-Initio Molecular Dynamics (AIMD), classical Molecular Dynamics (MD), and Finite Element Analysis (FEA) to investigate atomic and cell-level properties. The considered Li-CO₂ battery consists of a lithium metal anode, an ionic liquid electrolyte, and a carbon cloth cathode with Sb_{0.67}Bi_{1.33}Te₃ catalyst. DFT and AIMD determined the electrical conductivities of Sb_{0.67}Bi_{1.33}Te₃ and Li₂CO₃ using the Kubo-Greenwood formalism and studied the CO₂ reduction mechanism on the cathode catalyst. MD simulations calculated the CO₂ diffusion coefficient, Li⁺ transference number, ionic conductivity, and Li⁺ solvation structure. The FEA model, parameterized with atomistic simulations data, reproduced the available experimental voltage-capacity profile at 1 mA/cm² and revealed spatio-temporal variations in Li₂CO₃/C deposition, porosity, and CO₂ concentration dependence on discharge rates in the cathode. Accordingly, Li₂CO₃ can form large and thin film deposits, leading to dispersed and local porosity changes at 0.1 mA/cm² and 1 mA/cm², respectively. The capacity decreases exponentially from 81,570 mAh/g at 0.1 mA/cm² to 6,200 mAh/g at 1 mA/cm², due to pore clogging from excessive discharge product deposition that limits CO₂ transport to the cathode interior. Therefore, the performance of Li-CO₂ batteries can be improved by enhancing CO₂ transport, regulating Li₂CO₃ deposition, and optimizing cathode architecture.

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1. Introduction

Lithium-carbon dioxide (Li-CO₂) batteries, with their high theoretical energy density and CO₂ fixation capability, have emerged as a potential energy storage solution [1, 2]. These batteries provide a theoretical specific energy density of 1876 Wh kg⁻¹, which is substantially higher than that of current lithium-ion battery systems [3, 4]. This high energy density makes Li-CO₂ batteries particularly attractive for applications in aviation, aerospace, and electric vehicles, where lighter and more compact energy storage solutions are crucial [5]. The potential for electric vehicles with much longer ranges or considerably lighter batteries could revolutionize the automotive market [6, 5]. However, Li-CO₂ batteries are still in early development stages and face challenges including reaction reversibility issues, low energy efficiency, and the need for improved cathode designs [7]. Ongoing research and development efforts focus on overcoming these limitations, with recent advancements showing promising results [4, 8, 9, 10, 11, 12, 13, 14]. The Li-CO₂ batteries operate on a unique electrochemical principle involving the reversible reaction between lithium and carbon dioxide [1, 2] contributing to CO₂ capture and utilization efforts [11, 15]. During the discharge process, lithium oxidizes at the anode, releasing lithium ions and electrons ($Li \rightarrow Li^+ + e^-$). Besides, at the cathode in the presence of solid catalysts [16, 4], CO₂ undergoes reduction ($3CO_2 + 4e^- \rightarrow 2CO_3^{2-} + C$), combining with lithium ions and electrons to form primarily lithium carbonate (Li₂CO₃) and carbon (C) ($4Li^+ + 3CO_2 + 4e^- \rightarrow 2Li_2CO_3 + C$). This discharge reaction typically occurs at a voltage of approximately 2.8 V [17]. The charging process reverses these reactions, decomposing the discharge products to regenerate lithium and CO₂. However, due to the stability of Li₂CO₃, this process requires higher voltages, often exceeding 3.8 V [18]. This increased voltage requirement underscores one of several key challenges facing Li-CO₂ battery technology. In particular, the formation of solid Li₂CO₃ and carbon can lead to electrode clogging and increased internal resistance [9, 19]. Moreover, the high charging voltages needed to decompose Li₂CO₃ significantly reduce the overall energy efficiency of the system [20, 21]. To address these issues, effective cathode catalysts are critical for enhancing reaction kinetics and minimizing overpotentials in Li-CO₂ batteries [8, 22]. Nevertheless, unwanted side reactions, particularly electrolyte decomposition, can occur at these high voltages [7]. Furthermore, the inherent inertness of CO₂ compounds the challenge, as it makes electrochemical reduction difficult and adversely affects both discharge voltage and energy efficiency [23, 2, 15].

In this work, we consider a Li-CO₂ battery cell comprising a lithium anode, a hybrid ionic liquid electrolyte, and a porous cathode. The electrolyte features a unique composition of 1-ethyl-3-methylimidazolium tetrafluoroborate (EMIM-BF₄) and dimethyl sulfoxide (DMSO) in a 2:3 ratio, supplemented with zinc iodide (ZnI₂) and 1 M lithium bis(trifluoromethanesulfonyl)imide

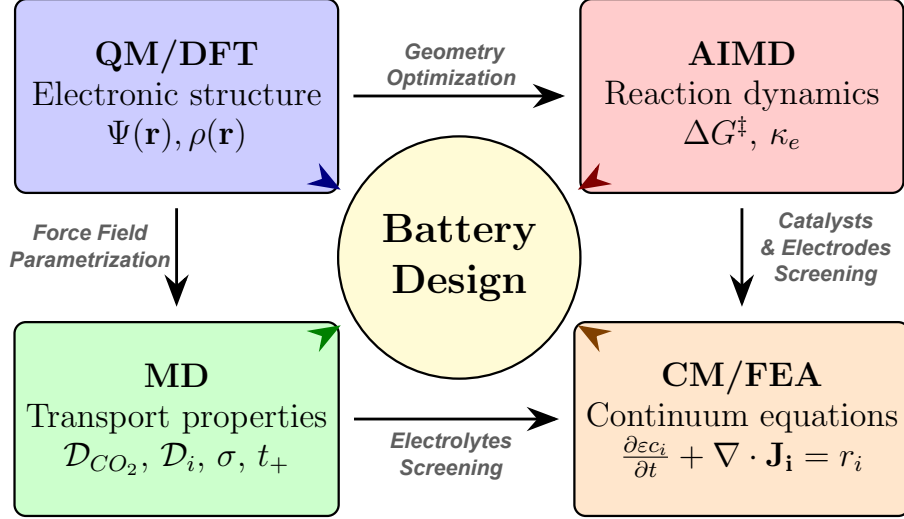


Figure 1: Diagram outlining the multiscale computational workflow for battery design, integrating four key methods: (1) quantum mechanics and Density Functional Theory calculations (QM/DFT), which calculate electronic structure and optimize atomic geometries; (2) Ab Initio Molecular Dynamics (AIMD), which simulates reaction dynamics to screen electrodes and catalysts; (3) Molecular Dynamics (MD), which models transport properties for electrolyte design; and (4) continuum modeling/Finite Element Analysis (CM/FEA), which develops and solves continuum equations to predict system-level performance. All methods feed into battery design, with arrows indicating critical data exchange.

(LiTFSI) salt. This optimized electrolyte formulation, experimentally validated for superior performance [4], enables efficient ion transport and stability. The cathode utilizes carbon cloth coated with a novel transition metal trichalcogenide alloy catalyst ($\text{Sb}_{0.67}\text{Bi}_{1.33}\text{Te}_3$), which significantly enhances CO_2 reduction/oxidation kinetics and cycling stability [4]. While recent advancements in Li- CO_2 batteries have focused on developing novel cathode materials and electrolytes [10, 24, 25, 11, 14, 13, 26], most computational studies have been limited to atomistic-level investigations. To address this gap, we introduce the first comprehensive multiscale framework that integrates quantum-scale simulations, Molecular Dynamics, and continuum modeling of Li- CO_2 batteries. This approach uniquely connects atomic-scale phenomena, such as catalyst surface reactions and ion transport, to macroscopic cell performance, enabling holistic optimization of Li- CO_2 battery systems. This strategy is also needed for other battery types [27, 28, 29, 30]. Our methodology advances the field by addressing critical knowledge gaps in the spatiotemporal evolution of parameters and the interdependencies between electrochemical processes at different scales, and it can be extended to the study of other battery systems.

At the Li- CO_2 battery cell level, Finite Element Analysis (FEA) can effectively simulate charge/discharge cycles, enabling macroscopic analysis of com-

plex processes within the battery cell. These processes include ion transport through porous electrodes, reaction kinetics, and their impact on cell performance [31, 32, 33]. However, FEA requires accurate electrode and electrolyte properties as input parameters, which can be derived from atomistic simulations. To bridge the gap between atomic and cell-level simulations and to calculate the necessary parameters for FEA, we employed a closed-loop workflow from the atomic to the cell level, as summarized in Figure 1. We employ quantum mechanics (QM) and Density Functional Theory (DFT) calculations to determine partial charges, equilibrated geometries, and Molecular Dynamics Force Field (FF) parameters for battery constituents using small systems (number of atoms < 1000). Ab Initio Molecular Dynamics (AIMD) calculations are used to determine the electrical conductivity of Li_2CO_3 and $\text{Sb}_{0.67}\text{Bi}_{1.33}\text{Te}_3$, based on DFT-equilibrated geometries and employing the Kubo-Greenwood formula [34, 35, 36]. AIMD is also used to study the surface reactions of $\text{Sb}_{0.67}\text{Bi}_{1.33}\text{Te}_3$ as a cathode catalyst material. Large-scale MD simulations are conducted to investigate the transport properties of the liquid electrolyte. Specifically, we calculate the diffusion coefficients of Li^+ and CO_2 , as well as the Li^+ transference number, by examining the mean square displacement (MSD) of the particles over time using the Einstein equation [37], and the ionic conductivity using the Green-Kubo formula [38]. The atomic-scale insights gained from DFT/AIMD and MD simulations are then integrated into our FEA model, enabling a comprehensive analysis of cell-level performance. Our FEA simulations investigate various aspects of battery cell behavior, including specific capacity, discharge behavior, and voltage-capacity profiles, considering factors such as CO_2 transport properties, cathode porosity, and the deposition of Li_2CO_3 and C.

The Li- CO_2 continuum model (CM) of the Li- CO_2 battery that we developed, supported by accurately calculated parameters from atomistic-level simulations, enables us to make predictions about porosity, deposited species quantities, and CO_2 concentration variations—parameters that are difficult to assess through experimental methods alone. Furthermore, it allows us to evaluate these properties as a function of time, providing dynamic insights into the battery’s behavior at the cell level. This hierarchical, cross-validated workflow not only bridges the longstanding gap between atomistic understanding and macroscopic battery behavior but also enables targeted material screening and iterative optimization of cell components. The resulting predictive capability accelerates the discovery and engineering of Li- CO_2 batteries with enhanced energy density, cycle life, and operational safety. Moreover, our methodology provides a generalizable template for the rational design of other advanced battery chemistries, where multiscale interactions govern performance and reliability. Ultimately, this multiscale computational strategy offers a powerful tool to guide experimental efforts, reduce development time, and unlock the full potential of emerging energy storage technologies.

2. Methods and computational details

2.1. Density Functional Theory Calculations for Structure Optimization

First-principles Density Functional Theory (DFT) calculations were performed using the Vienna Ab-initio Simulation Package (VASP) [39, 40] to optimize the crystal structures of $\text{Sb}_{0.67}\text{Bi}_{1.33}\text{Te}_3$ and Li_2CO_3 . The $\text{Sb}_{0.67}\text{Bi}_{1.33}\text{Te}_3$ structure was constructed based on Sb_2Te_3 from the Materials Project database [41], and multiple configurations were optimized to identify the most energetically favorable structure. The Li_2CO_3 structure was also obtained from the Materials Project database. For surface interaction studies, the Perdew-Burke-Ernzerhof (PBE) functional within the generalized gradient approximation (GGA) was used, and dispersion interactions were accounted for using the D3 van der Waals correction [42]. These calculations are crucial for accurately modeling CO_2 adsorption and reduction processes on the catalyst surface. Further computational details are provided in the Supplementary Information.

2.2. Ab Initio Molecular Dynamics (AIMD) and Kubo-Greenwood method for electrical conductivity calculation

Using optimized structures of $\text{Sb}_{0.67}\text{Bi}_{1.33}\text{Te}_3$ and Li_2CO_3 compounds from our DFT calculation, we carried out the Ab Initio Molecular Dynamics (AIMD) simulations at 300K up to 1ps with NVT ensemble and 1fs time step to select 10 different frames in the 1ps trajectory data. Subsequently, we generated wavefunctions ($\Psi_{i,k}$ where i and k are band and k-point indices, respectively) data for the selected frames using DFT calculations as explained by Knyazev et. al. [43, 34] to compute the electronic conductivity based on Kubo-Greenwood (KG) formula. During the calculation of wavefunction data from VASP, we included 100% more unoccupied bands than occupied bands for both $\text{Sb}_{0.67}\text{Bi}_{1.33}\text{Te}_3$ and Li_2CO_3 compounds to account for excited state wavefunctions and their corresponding energies. The dynamic electrical conductivity $\kappa(\omega) = \kappa_1(\omega) + i\kappa_2(\omega)$ is represented as a coefficient of the current density $\mathbf{J}(\omega)$ and electric field $\mathbf{E}(\omega)$:

$$\mathbf{J}(\omega) = [\kappa_1(\omega) + i\kappa_2(\omega)]\mathbf{E}(\omega) \quad (1)$$

where ω is the frequency of the external electric field $\mathbf{E}(\omega)$. The important component $\kappa_1(\omega)$ relates to the energy absorbed by the electrons and is computed by the Kubo-Greenwood formula as [34, 35, 36]

$$\kappa_1(\omega) = \frac{2\pi e^2 \hbar^2}{3m\omega\Omega} \sum_{i,j,\alpha,k} W(k) |\langle \Psi_{i,k} | \nabla_\alpha | \Psi_{j,k} \rangle|^2 [f(\varepsilon_{i,k}) - f(\varepsilon_{j,k})] \delta(\varepsilon_{j,k} - \varepsilon_{i,k} - \hbar\omega) \quad (2)$$

Here, the sum is over all $\mathbf{k}$ points in the Brillouin zone, band indices i and j and three spatial dimensions α . $\kappa_1(\omega)$ is the frequency-dependent electrical conductivity. e is the elementary charge. $\hbar$ is the reduced Planck constant. m is the electron mass. ω is the angular frequency. Ω is the volume of the system. f_i and f_j are the Fermi-Dirac distribution functions for states i and j . $|\psi_i\rangle$ and $|\psi_j\rangle$ are the wavefunctions of states i and j . $\varepsilon_{i,k}$ and $\varepsilon_{j,k}$ are

the energies of states i and j . δ is the Dirac delta function. $W(k)$ denotes a weight function at a particular $\mathbf{k}$ point. The wave function $\Psi_{i,\mathbf{k}}$ is generated from our AIMD simulations at room temperature as explained above. $f(\varepsilon_{i,\mathbf{k}})$ is the Fermi function. In practical calculations, δ -function is written in terms of the Gaussian function:

$$\delta(\varepsilon_{j,k} - \varepsilon_{i,k} - \hbar\omega) \rightarrow \frac{1}{\sqrt{2\pi}\Delta E} \exp\left(-\frac{(\varepsilon_{j,k} - \varepsilon_{i,k})^2}{2(\Delta E)^2}\right) \quad (3)$$

where ΔE is broadening of the δ -function. This technical parameter plays an essential role in controlling the algorithm in the Kubo-Greenwood formula, particularly at low frequencies.

2.3. Atomistic Molecular Dynamics Simulations (MD)

Classical Molecular Dynamics (MD) simulations were performed using LAMMPS [44] to investigate the electrolyte system relevant to Li-CO₂ batteries. The simulated electrolyte consisted of DMSO:(EMIM⁺, BF₄⁻) at a 3:2 ratio, 1 M (Li⁺, TFSI⁻), and included 6000 DMSO molecules, 1857 (EMIM⁺, BF₄⁻) pairs, 566 (Li⁺, TFSI⁻), 14 ZnI₂, and 14 CO₂ molecules, totaling 113,142 atoms. Initial configurations were generated and energy minimized, followed by agitation using the Langevin and Nosé-Hoover thermostats. The system was equilibrated to ambient conditions (300 K, 1 bar) in stages using the Berendsen barostat, and transport properties were evaluated under experimental-like conditions in the NVT ensemble at 300 K. Further details on force field parameterization, cutoff schemes, and equilibration protocols are provided in the Supplementary Information.

2.4. Finite element model

A mathematical framework has been developed for modeling the behavior of prismatic Li-CO₂ batteries, drawing inspiration from previous studies on Li-O₂ and Li-ion batteries [45, 46, 47]. The model integrates key principles such as mass and current conservation, species transport phenomena, and reaction kinetics within the cathode and separator regions to provide deeper insights into the complex mechanisms that occur within the cell during operation. The Li-CO₂ cell under investigation consists of several components: a lithium metal sheet serving as the negative electrode, a separator, a porous carbon cathode coated with Sb_{0.67}Bi_{1.33}Te₃, and a liquid electrolyte that permeates the porous structures, as illustrated schematically in Figure 2. The used electrolyte composition is described in section 2.3. To facilitate electron transfer, current collectors are located at the rear of both electrodes. This configuration enables a thorough examination of the electrochemical and transport processes that govern battery cell performance and limitations. The main objective of this mathematical framework is to provide a comprehensive understanding of the intricate interactions among various components and processes within the Li-CO₂ battery cell system.

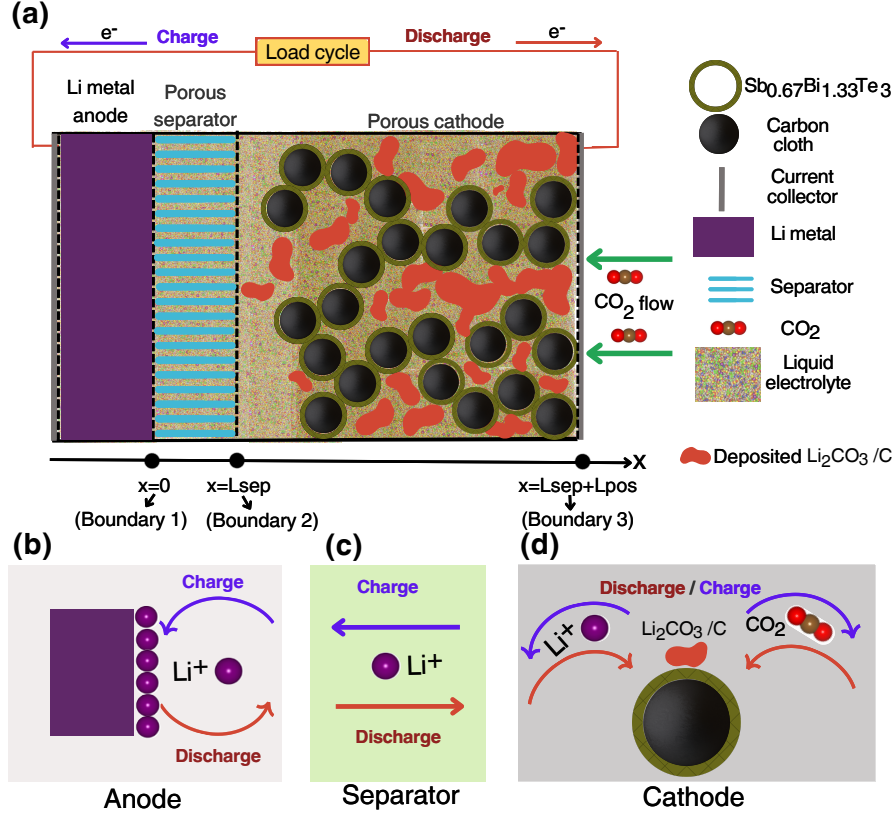


Figure 2: (a) Schematic diagram of the Li-CO₂ continuum model used in this study. The model consists of two active domains: the separator (from boundary 1 to boundary 2) and the porous electrode (from boundary 2 to boundary 3). The Li anode and CO₂ inlet are represented as boundaries 1 and 3, respectively. The inset at the bottom illustrates the key processes: (b) surface reactions at the anode, (c) ion transport within the separator, and (d) electrochemical reactions at the cathode.

2.4.1. Electrode Reactions

Cathode Reaction

The porous carbon electrode provides a site for the electrochemical reduction of carbon dioxide. During operation, CO₂ from the external feed side dissolves in the electrolyte, moves through the pores of the positive electrode, and reacts with lithium ions at the active sites. The reaction considered at the positive electrode is [11]:

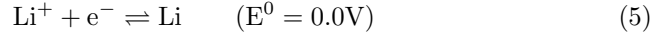


where $E_d^0 = 2.8\text{V}$ and $E_c^0 = 3.8\text{V}$ represent the equilibrium potentials for discharge and charge reactions, respectively [17, 18]. The reaction products

(Li_2CO_3 and C) have limited solubility in the DMSO-based liquid electrolyte (0 mol/L for C and 0.0371 mol/L for Li_2CO_3). When their concentrations exceed these solubility limits, the electrolyte solution becomes supersaturated [48], leading to the deposition of Li_2CO_3 and C. This is followed by the nucleation and growth of these products in the liquid phase within the porous electrode. These deposited products significantly impact the battery's performance by covering the active surface area, obstructing pathways for reactive species (Li^+ and CO_2), and inhibiting further reactions within the cathode, thereby limiting the extent of cell discharge.

Anode Reaction

The main reaction at the anode, which represents the reversible deposition and dissolution of lithium metal during battery cell operation, is described as [46]:



Where E^0 is the reaction equilibrium potential for charge and discharge [46].

2.4.2. Particle transport at the macroscopic level

The Nernst-Planck equation, a continuity equation for the time-dependent concentration $c_i(t, \mathbf{x})$ of a chemical species, is used to describe the transport of chemical species within the battery cell:

$$\frac{\partial \varepsilon c_i}{\partial t} + \nabla \cdot \mathbf{J}_i = r_i \quad (6)$$

In this context, ε represents the porosity of the electrode or the electrolyte region in the solution phase, $\mathbf{J}_i$ denotes the flux, and r_i indicates the volumetric production rate of species i transitioning from the electrode material to the electrolyte within the porous structure. The total flux is composed of three components: diffusion, advection, and electromigration. This implies that the concentration is affected by an ionic concentration gradient ∇c , flow velocity $\mathbf{v}$, and an electric field $\mathbf{E}$.

$$\mathbf{J} = - \underbrace{D^{eff} \nabla c}_{\text{Diffusion}} + \underbrace{c \mathbf{v}}_{\text{Advection}} + \underbrace{\frac{D^{eff} z e}{k_B T} c \mathbf{E}}_{\text{Electromigration}} \quad (7)$$

In this equation, D^{eff} denotes the effective diffusion coefficient of the chemical species, z represents the valence of the ionic species, e represents the elementary charge, k_B represents the Boltzmann constant, and T represents the absolute temperature. The electric field may be further decomposed as:

$$\mathbf{E} = -\nabla \phi - \frac{\partial \mathbf{A}}{\partial t} \quad (8)$$

where ϕ is the electric potential and A is the magnetic vector potential. Therefore, the Nernst–Planck equation is given by:

$$\frac{\partial c}{\partial t} = \nabla \cdot \left[D^{eff} \nabla c - c \mathbf{v} + \frac{D^{eff} z e}{k_B T} c \left(\nabla \phi + \frac{\partial \mathbf{A}}{\partial t} \right) \right] \quad (9)$$

The primary transport mechanisms in batteries, ionic migration due to electric fields and diffusion due to concentration gradients, are accurately described by the Nernst–Planck equation, which excludes time-varying magnetic effects [49]:

$$\frac{\partial c}{\partial t} = \nabla \cdot \left[D^{eff} \nabla c - c \mathbf{v} + \frac{D^{eff} z e}{k_B T} c \nabla \phi \right] \quad (10)$$

The moving boundaries theory of transference number (t_+) gives [50]:

$$t_+ = \frac{v c_+ F}{I \Delta t} \quad (11)$$

where v is the volume occupied by diffusing cations by the boundary in time Δt , c_+ the Li^+ concentration, F the Faraday constant, and I the electric current. By introducing the electrode surface S :

$$t_+ = \frac{v c_+ F / S}{I \Delta t / S} \quad (12)$$

Also, we can write c as:

$$c = \frac{N}{N_A v} \quad (13)$$

with N the number of particles and N_A the Avogadro number.

Then:

$$t_+ = \frac{\mathbf{J} F}{i_2} \quad (14)$$

with, $\mathbf{J}$ the molar flux of cations, $i_2 = I/S$ the current density in the electrolyte, and F the Faraday constant.

The primary interest in battery modeling is electrochemical processes rather than fluid dynamics, so the advection term is considered less critical [52]. Furthermore, the porous structure limits large-scale fluid motion, further reducing the importance of advection. By neglecting the Advection term, the Nernst–Planck equation for Li^+ and CO_2 can be expressed as:

$$\mathbf{J}_{\text{Li}^+} = -D_{\text{Li}^+}^{eff} \nabla c_{\text{Li}^+} + \frac{i_2 t_+}{F} \quad (15)$$

$$\mathbf{J}_{\text{CO}_2} = -D_{\text{CO}_2}^{eff} \nabla c_{\text{CO}_2} \quad (16)$$

The effective diffusion coefficient of a species i ($i=\text{Li}^+$, CO_2) was calculated using the Bruggeman equation [32, 33]:

$$D_i^{eff} = \varepsilon^{1.5} D_i \quad (17)$$

Table 1: Parameters used in the Finite Element Analysis (FEA).

Symbol	Value	Description	Reference
Parameters used in the battery assembly			
L_{sep}	50 μ m	Thickness of Separator	This work
L_{pos}	2 mm	Thickness of positive electrode	This work
A_c	1 cm ²	Cross-sectional area	This work
$\varepsilon_{l,0}$	0.73	Initial porosity of positive electrode	This work
$m_{Sb_{0.67}Bi_{1.33}Te_3}$	0.4 mg	mass of active cathode material	This work
ε_{sep}	0.5	Porosity of separator	
$\varepsilon_{s,0}$	$1 - \varepsilon_{l,0}$	Initial active material solid fraction of cathode	This work
$c_{Li,0}$	1000 mol/m ³	Initial concentration of Li ⁺ in electrolyte	This work
Parameters calculated using DFT/AIMD simulation			
κ_{pos}	1.9 10 ⁵ S/m	Conductivity of positive electrode	This work
$\kappa_{Li_2CO_3}$	1.8 10 ⁻⁹ S/m	electrical conductivity of Li ₂ CO ₃ film	This work
Parameters calculated using MD simulation			
D_{Li^+}	3.17 10 ⁻⁹ m ² /s	Diffusion coefficient of Li ⁺ in electrolyte	This work
D_{CO_2}	1.2 10 ⁻⁹ m ² /s	CO ₂ diffusion coefficient	This work
σ	6.77 mS/cm	Conductivity of Li ⁺ in electrolyte	This work
t_+	0.91	Transference number of Li ⁺ in electrolyte	This work
General parameters			
$i_{0,Li}$	0.965 A/m ²	Exchange current density for anode	[51, 46]
$\text{dln}f_{\pm}/\text{dln}c$	-1.03	Activity dependence	[51, 46]
E_d^0	2.8 V	Equilibrium potential for discharge	[17]
E_c^0	3.8 V	Equilibrium potential for charge	[18]
$r_{pos,0}$	25 nm	Particle radius in the positive electrode	[46]
a_0	$3\varepsilon_{s,0}/r_{pos,0}$	Initial active specific surface area	[46]
n	4	Number of transferred electrons	[17]
$c_{Li_2CO_3,max}$	0.0371 mol/L	Solubility limit of Li ₂ CO ₃ in DMSO	
$c_{C,max}$	0 mol/L	Solubility limit of Carbon in DMSO	
$\rho_{Li_2CO_3}$	2140 kg/m ³	Density of Li ₂ CO ₃	
$M_{Li_2CO_3}$	73.89 10 ⁻³ kg/mol	Molecular weight of Li ₂ CO ₃	
ρ_C	2.11 g/cm ³	Density of carbon	
T	300 K	Temperature	
Assumed parameters			
k_a	1.11 10 ⁻¹⁵ m/s	Anodic reaction rate coefficient	Assumed
k_c	3.14 10 ⁻¹⁵ m ⁷ /s/mol ²	Cathodic reaction rate coefficient	Assumed
$c_{CO_2,0}$	4.8 mol/m ³	Initial CO ₂ concentration in positive electrode	Assumed

The $\varepsilon^{1.5}$ form, in the Bruggeman relation, emerges from the mathematical treatment of a system of spherical particles dispersed in a continuous medium [53]. This form has been found to provide a reasonable empirical fit approximation for many real porous systems, especially at higher porosities [53, 54]. The diffusion coefficient D_i ($i=Li^+$, CO₂) is calculated from our MD simulation.

Table 2: Boundary conditions at boundaries 1 ($x=0$), 2 ($x=L_{sep}$), and 3 ($x=L_{sep} + L_{pos}$).

Variable	Value	Boundary
Total current density	i_{app}	3
c_{CO_2}	$c_{CO_2,0}$	3
ϕ_1	0 V	1
∇c_{CO_2}	0	2
∇c_{Li}	0	1, 3

Table 3: Initial values of independent and dependent variables.

Variable	Value	Domain
ϕ_1	2.8 V	Porous cathode
ϕ_1	0 V	Separator
ϕ_2	0 V	Porous cathode, Separator
c_{Li}	$c_{Li,0}$	Porous cathode, Separator
c_{CO_2}	$c_{CO_2,0}$	Porous cathode
$c_{Li_2CO_3}$	$c_{Li_2CO_3,max}$	Porous cathode
c_C	$c_{C,max}$	Porous cathode
ε	$\varepsilon_{l,0}$	Porous cathode

2.4.3. Equations of battery electrochemical properties

The surface density of the liquid phase $\mathbf{i}_2$ represents the function of the electrolyte concentration c_{Li} , time t , the ionic conductivity σ^{eff} , the absolute temperature T , the activity coefficient $\frac{\partial \ln f_{\pm}}{\partial \ln c_{Li}}$, the universal gas constant R , the cathode porosity ε and electrode solution-phase potential ϕ_2 [45, 46].

$$\mathbf{i}_2 = -\sigma^{eff} \nabla \phi_2 + \left(\frac{2\sigma^{eff} RT}{F} + \frac{2\sigma^{eff} RT}{F} \frac{\partial \ln f_{\pm}}{\partial \ln c_{Li}} \right) (1 - t_+) \nabla \ln c_{Li} \quad (18)$$

$$\sigma^{eff} = \varepsilon^{1.5} \sigma \quad (19)$$

The solid phase current density $\mathbf{i}_1$ is the function of the solid-electrode potential ϕ_1 and the electronic conductivity of solid-phase mixture κ_{pos} [45, 46],

$$\mathbf{i}_1 = -\kappa_{pos}^{eff} \nabla \phi_1 \quad (20)$$

$$\kappa_{pos}^{eff} = (1 - \varepsilon_{l,0})^{1.5} \kappa_{pos} \quad (21)$$

$\varepsilon_{l,0}$ denotes the starting porosity of the cathode. It is important to mention that the effect of deposits (Li_2CO_3 and C) with low conductivity on κ_{pos} was not taken into account, and it was assumed that κ_{pos} would not change even as the battery reactions occurred. This assumption is primarily justified by the demonstrated structural and chemical stability of the catalyst under cycling conditions [4]. In our continuum model, we further decouple electron transport through the deposited Li_2CO_3 and carbon byproducts, treating their intrinsic

conductivities separately from that of the catalyst. This approach ensures that the catalyst's role in maintaining electronic percolation pathways is accurately represented. However, it is important to acknowledge the potential limitations of this simplification. If the catalyst were not stable—for example, if it were susceptible to pulverization or surface passivation—its conductivity could decrease during cycling. Such changes might result in localized current hotspots or capacity fade, which would not be captured by the current model. Therefore, while the assumption is reasonable for stable catalyst systems, it may not be universally applicable to all materials or operational conditions.

Conservation of charge

$$\nabla \mathbf{i}_1 + \nabla \mathbf{i}_2 = 0 \quad (22)$$

$$\sum_i s_i M_i^{z_i} \rightarrow n e^- \quad (23)$$

$$\nabla \cdot \mathbf{i}_2 = a j \quad (24)$$

These equations demonstrate that the transfer current per unit electrode volume corresponds to the electrode chemical reaction rate, where M_i is a symbol representing a species participating in the electrochemical reaction, z_i and s_i are the charge number and the stoichiometric coefficient of species i , respectively, n is the number of electrons transferred in the reaction, a is the specific interfacial area of the pore per unit electrode volume, and j is the average transfer current. In the present work, only one electrochemical reaction is considered: the formation of $\text{Li}_2\text{CO}_3/\text{carbon}$ ($\text{Li}_2\text{CO}_3/\text{C}$) inside the porous cathode. In practical cells, battery side reactions can occur during cycling alongside the main electrode reactions. This is reflected by the superficial production rate r_i of a species from the cathode's solid phase to the pore solution, given by Faraday's law as a function of the local transfer current density j_c between the cathode/electrolyte interface.

$$r_i = -\frac{a s_i}{n F} j_c \quad (25)$$

Rate expressions

For modeling the cathode's electrochemical reaction, a modified version of the Butler-Volmer equation is applied in the model using two rate coefficients. This modification is necessary because the cathode reaction presented in equation 4 depends on both the concentrations of Li^+ and CO_2 , as well as the concentration of $\text{Li}_2\text{CO}_3/\text{C}$ [55]:

$$\frac{j_c}{n F} = k_a (c_{\text{Li}_2\text{CO}_3/\text{C},s}) \exp \left[\frac{(1-\beta)nF}{RT} \eta_c \right] - k_c (c_{\text{Li}^+,s})^2 (c_{\text{CO}_2,s}) \exp \left[\frac{-\beta n F}{RT} \eta_c \right] \quad (26)$$

$$\eta_c = \phi_1 - \phi_2 - \Delta \phi_{film} - E^0 \quad (27)$$

$$\phi_{film} = j_c R_{film} \varepsilon_s \quad (28)$$

where j_c is local transfer current density between electrode and electrolyte interface, $c_{i,s}$ is the molar concentration of species i at the cathode surface, k_a and k_c are the anodic and cathodic rate constant, respectively, β is a symmetry factor equal to 0.5, η_c is surface or activated overpotential for reaction at cathode, ϕ_{film} and R_{film} are the voltage drop and the electrical resistivity across $\text{Li}_2\text{CO}_3/\text{C}$ film formation, respectively, ε_s is the volume fraction of solid $\text{Li}_2\text{CO}_3/\text{C}$, and E^0 is the theoretical open-circuit potential for reaction.

The properties of the $\text{Li}_2\text{CO}_3/\text{C}$ film, specifically its electrical resistivity R_{film} , thickness (Δs), and conductivity (κ_{film}), are related by the equations below:

$$R_{film} = \frac{\Delta s}{\kappa_{film}} \quad (29)$$

$$\Delta s = \Delta s_C + \Delta s_{\text{Li}_2\text{CO}_3} \quad (30)$$

$$\Delta s_i = \frac{\Delta \varepsilon_i}{a} = \frac{M_i c_i}{a \rho_i} \quad (31)$$

where i represents Li_2CO_3 or C, while Δs_i , $\Delta \varepsilon_i$, M_i , a , and ρ_i refer to the film thickness of i , the change in porosity of the cathode caused by i , the molar mass of i , the specific area, and the density of i , respectively.

The electrochemical reaction at the lithium metal anode is given by the general Butler-Volmer equation as follows [45, 46]:

$$j_a = i_0 \exp \left[\frac{(1 - \beta)nF}{RT} \eta_a \right] - \exp \left[\frac{-\beta nF}{RT} \eta_a \right] \quad (32)$$

where i_0 is the exchange current density for the anode, η_a is the surface or activated overpotential for the reaction at the anode, and the other parameters are as described above.

The specific area a of the electrode/electrolyte interface in Equation 31 is decreased by the morphology and dynamic change in porosity due to $\text{Li}_2\text{CO}_3/\text{C}$ solid passivation during the electrochemical reaction. This effective local surface area per unit volume of the electrode can be commonly written by a geometric relation [45, 46]:

$$a = a_0 \exp \left[1 - \frac{\varepsilon_{\text{Li}_2\text{CO}_3/\text{C}}}{\varepsilon_{l,0}} \right] \quad (33)$$

where a_0 the initial specific surface area, $\varepsilon_{\text{Li}_2\text{CO}_3/\text{C}}$ and $\varepsilon_{l,0}$ are the volume fraction of solid $\text{Li}_2\text{CO}_3/\text{C}$ and initial electrode porosity, respectively.

Transport through $\text{Li}_2\text{CO}_3/\text{C}$ layer

During discharge, the formed $\text{Li}_2\text{CO}_3/\text{C}$ creates an additional barrier for Li^+ and CO_2 diffusion from the porous medium to the cathode surface.

This process follows Fick's law [46], where the flux is proportional to the concentration gradient between the bulk electrolyte and the active surface, sustained by ongoing electrochemical reactions consuming reactants.

$$-\frac{as_i}{nF}j_c = \frac{aD_{i,fil}^{eff}(c_i - c_{i,s})}{l} \quad (34)$$

where $D_{i,fil}^{eff}$ is the effective diffusion coefficient of species i across the $\text{Li}_2\text{CO}_3/\text{C}$ film, and l is the thickness of the film. c_i represents the molar concentration of species i in the electrolyte, while $c_{i,s}$ denotes the molar concentration of species i specifically at the cathode surface. The effective diffusivity for both Li^+ and CO_2 reactants is used to describe how the pores are deposited by the $\text{Li}_2\text{CO}_3/\text{C}$ formation; this can be described by the Bruggeman relation [32, 33]:

$$-\frac{\partial \varepsilon}{\partial t} = \frac{aj_c M_{\text{Li}_2\text{CO}_3/\text{C}}}{2F \rho_{\text{Li}_2\text{CO}_3/\text{C}}} \quad (35)$$

The volume fraction of the solid $\text{Li}_2\text{CO}_3/\text{C}$ and the thickness of the film (l) can be determined from the cathode volume balance as:

$$\varepsilon_{\text{Li}_2\text{CO}_3/\text{C}} = 1 - \varepsilon - \varepsilon_{l,0} \quad (36)$$

$$l = \left[\frac{\varepsilon_{\text{Li}_2\text{CO}_3/\text{C}} + \varepsilon_{s,0}}{\varepsilon_{s,0}} \right]^{(1/3)} r_0 - r_0 \quad (37)$$

In this context, $M_{\text{Li}_2\text{CO}_3/\text{C}}$ and $\rho_{\text{Li}_2\text{CO}_3/\text{C}}$ represent the molar mass and density of $\text{Li}_2\text{CO}_3/\text{C}$, respectively. The term $\varepsilon_{s,0}$ denotes the volume fraction of the initial solid phase of the cathode material, while r_0 indicates the radius of the particles within the electrode.

Initial conditions/Boundary conditions

The governing equations consist of partial differential equations that include time and space as independent variables. The dependent variables were simulated using COMSOL Multiphysics 6 Finite Element Analysis software to examine their spatial and temporal changes. The battery cell was segmented into three regions: the anode, separator, and porous cathode. An extra fine mesh was selected for enhanced accuracy. To reduce computation time, a variable-step solver was employed with a relative tolerance of 1×10^{-4} . This solver decreases the total number of time steps required for the calculation. The parameters used in the simulations are presented in Tables 1, 2, and 3. The applied current density, i_{app} , was considered positive when directed from the cathode to the external circuit (Figure 2). The total current density passing through boundary 3 was designated as i_{app} . The Li^+ flux at boundaries 1 and 3 (J_{Li}) and CO_2 flux at boundary 2 (J_{CO_2}) were set to zero. Additionally, the boundary condition $\nabla c_i = 0$ was applied to the relevant boundaries (see Table 2). The parameter c_{CO_2} was assumed to be equal to the saturation concentration, $c_{\text{CO}_2,0}$, at boundary 3. The boundary conditions are outlined in Table 2. Discharge simulations were conducted with applied current densities (i_{app}) ranging from 0.1 to 1 mA/cm^2 , defined as positive in the direction from the cathode to the external circuit.

3. Results and discussion

3.1. Electrical conductivity from DFT, AIMD, and Kubo-Greenwood formula

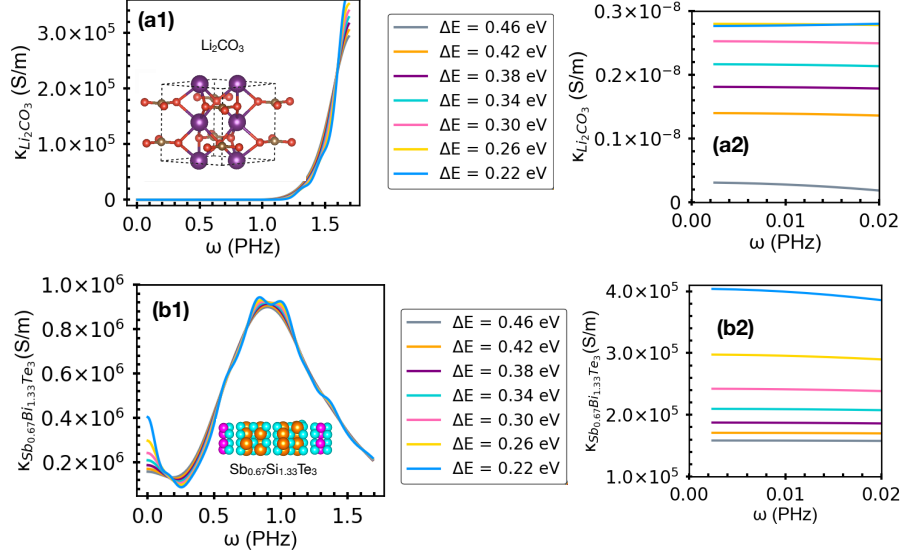


Figure 3: Dynamic electrical conductivity of Li_2CO_3 (a1-a2) and $\text{Sb}_{0.67}\text{Bi}_{1.33}\text{Te}_3$ (b1-b2) calculated using the Kubo-Greenwood formula for a wide frequency range (a1 and b1) and a small frequency range (a2 and b2). The electrical conductivity is plotted using ΔE values from 0.22 to 0.46 eV.

The calculated dynamic electrical conductivities for $\text{Sb}_{0.67}\text{Bi}_{1.33}\text{Te}_3$ and Li_2CO_3 are shown in Figure 3 (a1) and (b1) across a broad frequency range, and in Figures 3 (a2) and (b2), which focus on low frequencies to enable extrapolation to 0 Hz, with ΔE values ranging from 0.22 eV to 0.46 eV. Static electrical conductivities at zero frequency were determined by extrapolating the low-frequency data in Figures 3 (a2) and (b2). The calculated conductivities are within the experimental order of magnitude. ΔE values between 0.34 eV and 0.42 eV were selected because they show convergence with experimental conductivity data within this range. Furthermore, varying the electrical conductivity within the computational uncertainty does not significantly affect the predicted battery properties in the FEA. The static electrical conductivities calculated for different values of ΔE show the conductive behavior of $\text{Sb}_{0.67}\text{Bi}_{1.33}\text{Te}_3$ (a conductivity of the order of 10^5 S/m) and insulative behavior of Li_2CO_3 (a conductivity of the order of 10^{-9} S/m). Our computational analysis reveals a static electrical conductivity of $(1.9 \pm 0.3) \times 10^5$ S/m for $\text{Sb}_{0.67}\text{Bi}_{1.33}\text{Te}_3$, indicating its strong electrical conducting properties. This finding aligns with the well-established characteristics of bismuth telluride (Bi_2Te_3) and its alloys, which are extensively employed in thermoelectric devices due to their good electrical

conductivity [56, 57]. The high conductivity can be attributed to the semiconductor nature of Bi_2Te_3 and the antimony (Sb) doping effect, which augments the carrier concentration [58]. Experimental investigations on Bi_2Te_3 have reported electrical conductivity values as high as $1.71 \times 10^5 \text{ S/m}$ at 323 K [58], further corroborating the computational results. Our calculations reveal that Li_2CO_3 exhibits a static electrical conductivity of $(1.8 \pm 0.3) \times 10^{-9} \text{ S/m}$ at 300 K, confirming its highly insulating nature. This result closely aligns with experimental measurements of $8 \times 10^{-9} \text{ S/m}$ at 393 K [59]. The slight difference between calculated and experimental values is consistent with the expected increase in conductivity at higher temperatures, underscoring the reliability of our computational approach. The insulating nature of Li_2CO_3 can be attributed to its ionic nature and the strong ionic bonds that hinder electron mobility [60]. In Li_2CO_3 , Li^+ ions act as the primary charge carriers rather than electrons, a characteristic typical of ionic compounds. The low conductivity of Li_2CO_3 presents challenges to battery performance, including high internal resistance and passivation of the electrode surface [61, 62]. The calculated electrical conductivities are used in the continuum model to characterize electron percolation in the cathode and the deposited Li_2CO_3 during the battery charge/discharge cycle.

3.2. Surface reaction of $\text{Sb}_{0.67}\text{Bi}_{1.33}\text{Te}_3$ as cathode catalyst

To comprehensively understand the formation of Li_2CO_3 and C on the $\text{Sb}_{0.67}\text{Bi}_{1.33}\text{Te}_3$ catalyst surface, we calculated the Gibbs free energy (ΔG) changes for various elementary reaction steps on this surface as shown below (steps 1-5) similar to the calculations done in Ref. [4]. Here, we used a slightly bigger supercell (as shown in Figure 3 (a1)) compared to the work done in Ref. [4]. The reaction pathway revealed a sequence of intermediate stages, beginning with the adsorption of CO_2^* on a Bi site with a Bi-O distance of 3.5 Å, which was marginally endergonic ($\Delta G = 0.002 \text{ eV}$). The coupled addition of Li^+ and an electron to the adsorbed CO_2^* formed the LiCO_2^* intermediate, a highly exergonic step ($\Delta G = -2.3 \text{ eV}$). Subsequent interactions between two LiCO_2^* intermediates produced Li_2CO_2^* and CO_2^* , with a ΔG of -1.2 eV . Further, the conversion of Li_2CO_2^* and CO_2^* into Li_2CO_3 and CO^* ($\Delta G = -1.7 \text{ eV}$) marked a pivotal step in the pathway. Finally, the generation of carbon was facilitated through the reaction of CO and CO_2 , an energetically uphill step ($\Delta G = 2.8 \text{ eV}$) that also produced CO_3^* on the surface. These results, illustrated in Figure 4, highlight the essential role of coupled electron-cation transfer processes in the growth of discharge products. The results in this paper are consistent with the work reported in Ref. [4]. The Gibbs free energy G of the reaction was calculated using the following equation:

$$G = E + \text{ZPE} + \int_0^T C_p dT - TS \quad (38)$$

where ZPE is the zero-point energy, C_p is the heat capacity, S is the entropy, T is the temperature, and E is the electronic energy.

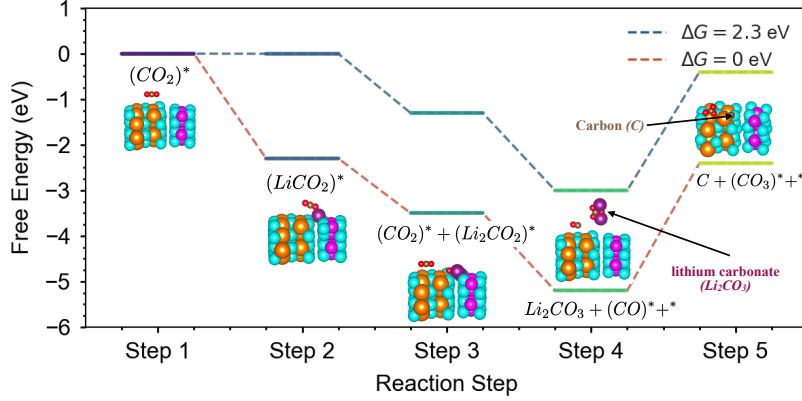
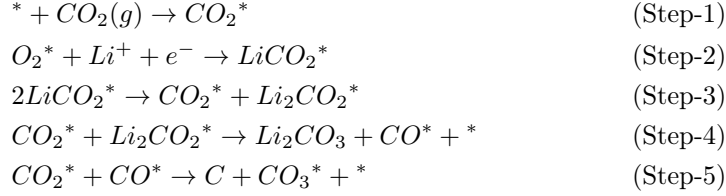


Figure 4: Free energy landscape of the CO_2 reduction reaction leading to the formation of $\text{Li}_2\text{CO}_3/\text{C}$ deposits on the $\text{Sb}_{0.67}\text{Bi}_{1.33}\text{Te}_3$ surface, with inset images illustrating the optimized geometries of key reaction intermediates.

In the calculation of Gibbs free energy profiles, full corrections including zero-point energy (ZPE) and entropy were applied explicitly to the initial gas-to-surface adsorption step of CO_2 , where such contributions are known to be significant due to the loss of translational and rotational entropy. For subsequent steps involving transformations between adsorbed intermediates (e.g., Li_2CO_3 , Li_2CO_2 , Li_2CO_3 , and $\text{C}-\text{CO}_3$ species), we assumed that ZPE and entropy effects largely cancel out, as these involve similar bonding environments and restricted surface-bound species. This approach is widely used in heterogeneous catalysis modeling and has been shown to introduce minimal quantitative error, while preserving the qualitative accuracy of energy landscapes and mechanistic conclusions [63]. Accordingly, our analysis emphasizes the dominant electronic energy contributions for these surface processes, which govern the reaction thermodynamics and barrier-lowering effects central to catalyst performance. The potential-dependent reaction energies were calculated based on the computational hydrogen electrode (CHE) model with reference to the Li/Li^+ electrode [64, 65]. The potential-dependent reaction energies profile is shown in Figure 4, and the respective reaction energies of each step are shown in Table 4. To level up the electrochemical step, a potential was applied to the formation of LiCO_2 . As can be seen in Figure 4, this potential occurs at 2.3 V versus Li/Li^+ electrode [4].

The complete 5-step reaction pathway can be summarized as follows:



The strong exergonic nature of steps 2-4 suggests that these processes occur readily, while the endergonic nature of the final carbon formation step aligns with the observed preferential adsorption of carbon on the $\text{Sb}_{0.67}\text{Bi}_{1.33}\text{Te}_3$ catalyst surface. The Li_2CO_3 is formed in step 4, but this step also releases CO^* and a free surface site ($*$). This suggests that Li_2CO_3 formation occurs near, but not directly on, the catalyst surface, reducing the cathode porosity and possibly clogging the cathode pores.

These insights into the reaction mechanism have significant implications for catalyst design and overall battery performance. The strong affinity of carbon for the catalyst surface, coupled with the energetically unfavorable step of formation, suggests that catalyst degradation through carbon accumulation could be a concern over long-term battery operation. This highlights the need for strategies to mitigate carbon build-up, such as developing self-cleaning catalysts or implementing periodic regeneration protocols. Furthermore, the spatial separation of Li_2CO_3 formation from the immediate catalyst surface underscores the importance of electrolyte engineering in Li- CO_2 battery design. Since Li_2CO_3 formation appears to occur away from the catalyst, the electrolyte's ability to facilitate Li_2CO_3 formation, dissolution, and transport becomes crucial.

The continuum model considers the formation of Li_2CO_3 and carbon in the cathode. We assume that the catalyst increases the reaction rate of $\text{Li}_2\text{CO}_3/\text{C}$ formation and use a reaction rate constant on the order of $10^{-15} \text{ m}^7 \text{ s}^{-1} \text{ mol}^{-2}$. This value is higher than the typical value assumed for Li_2O_2 formation in Li- O_2 batteries with non-catalyzed carbon cathodes, which is on the order of $10^{-17} \text{ m}^7 \text{ s}^{-1} \text{ mol}^{-2}$ [46].

Table 4: Reaction energies of each of the elementary steps.

Reaction Step	Reaction Steps	ΔG (eV)
Step-1	$\Delta G = G(\text{CO}_2^*) - G(*) - G(\text{CO}_{2,gas})$	0.002
Step-2	$\Delta G \approx \Delta E = E(\text{LiCO}_2^*) - E(\text{Li}^+ + e^-) - E(\text{CO}_2^*)$	-2.3
Step-3	$\Delta G \approx \Delta E = E(\text{Li}_2\text{CO}_2^*) + E(\text{CO}_2^*) - 2E(\text{LiCO}_2^*)$	-1.2
Step-4	$\Delta G \approx \Delta E = E(\text{Li}_2\text{CO}_3) + E(*) + E(\text{CO}^*) - E(\text{Li}_2\text{CO}_2^*) - E(\text{CO}_2^*)$	-1.7
Step-5	$\Delta G \approx \Delta E = E(\text{C}) + E(\text{CO}_3^*) + E(*) - E(\text{CO}^*) - E(\text{CO}_2^*)$	2.8

3.3. *Li-CO₂ battery electrolyte transport properties and Li⁺ solvation structure from large scale MD simulation.*

This section presents the results of MD simulations that elucidate the atomic-level dynamic and solvation properties of the Li-CO₂ battery electrolyte depicted in Figure 5 (a). These simulations provide microscopic insights into the electrolyte's behavior, bridging the gap between the atomic and macroscopic transport properties in our multiscale approach.

3.3.1. *Li-CO₂ battery electrolyte transport properties*

The primary goal is to investigate the electrolyte's dynamic properties governing ion and CO₂ transport mechanisms, including ionic conductivity, cation transference number, and CO₂ diffusion coefficient, as presented in Table 5. These properties directly impact the Li-CO₂ battery's electrochemical performance.

The cations' transference number is assessed by [66]:

$$t_+ = \frac{D_{\text{Li}^+}}{D_{\text{Li}^+} + D_{\text{TFSI}^-}} \quad (39)$$

Where the self-diffusion coefficients are determined from the mean-square-displacement of the center of mass presented in Figure 5 (b), calculated from the MD simulations as follows [67]:

$$\Delta^2 \mathbf{r}_{CM}(t) = \langle \Delta^2 \mathbf{r}(t) \rangle = \frac{1}{N} \sum_{i=1}^N [\mathbf{r}_i(t+t_0) - \mathbf{r}_i(t_0)]^2 \quad (40)$$

In the expression above, $\mathbf{r}_i(t)$ represents the temporal position of a random walker (ion i or CO₂ molecule). Here, t indicates the time, and t_0 is the initial time at which the random walker begins to move. The diffusion coefficients are extrapolated from the normal diffusion regime characterized by:

$$\langle \Delta^2 r \rangle = 6Dt, \quad t > \tau \quad (41)$$

where τ is the crossover time to the normal diffusion regime.

The Green-Kubo (GK) approach [38] that takes the autocorrelation with respect to the ionic current in the electrolyte $\mathbf{J}$ is used to calculate the ionic conductivity σ :

$$\sigma = \frac{V}{k_B T} \int_0^\infty dt \langle \mathbf{J}(t) \cdot \mathbf{J}(0) \rangle \quad (42)$$

with,

$$\mathbf{J}(t) = q \sum_i^N z_i \mathbf{v}_i(t) \quad (43)$$

Here q is the elementary charge, z_i represents the charge number (valence) of ion i , $\mathbf{v}_i$ is the velocity of ion i , T is temperature, k_B is the Boltzmann constant, V is the volume of the simulation box, and N is the number of ions.

Table 5: Transport Properties of [EMIM-BF₄:DMSO (2:3) + 1M LiTFSI + ZnI₂ + CO₂] electrolyte from Molecular Dynamics simulation.

Parameter	Value	Unit	Symbol
Diffusion coefficient of Li ⁺	3.17	10 ⁻⁹ m ² s ⁻¹	D _{Li⁺}
Ionic conductivity of the electrolyte	6.77 (7.4) ^a	mS cm ⁻¹	σ
Transference number of Li ⁺	0.91 (0.88) ^a	—	t ₊
Diffusion coefficient of CO ₂	1.2 (1.16) ^b (1.14) ^c	10 ⁻⁹ m ² s ⁻¹	D _{CO₂}

^a Results for 1M LiTFSI/DMSO electrolyte [68]

^b SEGWE Equation [69]

^c Einstein equation [69]

The estimated diffusion coefficient of CO₂ in DMSO at 25 °C is approximately 1.16×10^{-9} m²/s using the Stokes-Einstein-Gierer-Wirtz (SEGWE) relation and 1.14×10^{-9} m²/s using the Einstein equation [69]. These values are consistent with our MD calculation result of 1.2×10^{-9} m²/s presented in Table 5, further validating our findings.

The calculated diffusion coefficients, transference number, and ionic conductivity are directly incorporated into the FEA component of our multiscale modeling framework. These atomistically derived transport properties serve as critical input parameters for the FEA, enabling us to accurately simulate the intercalation and transport of CO₂ and ions during battery charge/discharge cycling at the cell level.

3.3.2. Li⁺ solvation structure from large-scale MD simulation

The structural analysis was conducted utilizing the Radial Distribution Function (RDF), g(r), and Coordination Number (Nc). The RDF quantifies the spatial distribution of particles relative to a reference particle, providing insights into the local structural organization that affects the transport properties in the electrolyte.

The RDF is mathematically expressed as:

$$g_{\alpha\beta}(r) = \frac{\langle \rho_{\beta}(r) \rangle}{\langle \rho_{\beta} \rangle_{local}} = \frac{1}{\langle \rho_{\beta} \rangle_{local}} \frac{1}{N_{\alpha}} \sum_{i \in \alpha} \sum_{j \in \beta} \frac{\delta(r_{ij} - r)}{4\pi r^2} \quad (44)$$

with $\langle \rho_{\beta}(r) \rangle$ the particle density of type β at a distance r around particles α , and $\langle \rho_{\beta} \rangle_{local}$ the particle density of type β averaged over all spheres around particles α with radius r_{max} . Here, we chose $r_{max}=12\text{\AA}$.

The primary peak in g(r) corresponds to the nearest neighbors, indicating the most probable interparticle distance and interaction strength. This relationship is described by $g(r) = e^{(-\beta U(r))}$, where $\beta = 1/(k_B T)$, and $U(r)$ is the interaction potential. Subsequent peaks in the RDF elucidate higher-order neighbor relationships, offering insights into medium-range order, coordination numbers, and packing efficiency. In isotropic, homogeneous systems, g(r) asymptotically approaches unity at large distances, signifying a uniform particle distribution.

The coordination number, N_c , is defined as the number of neighboring particles within a specified cutoff distance of the pair interaction potential from a central particle. It is mathematically represented as:

$$N_c = 2\pi \int_0^{r_c} n_r r dr = 2\pi n_b \int_0^{r_c} g(r) r dr \quad (45)$$

where r_c denotes the distance at which the interparticle interaction potential approaches zero, n_b represents the bulk density, and n_r is the average number density of particles at a given distance r , related to the RDF by $n_r = n_b g(r)$.

The coordination number as a function of distance, $N(r)$, is expressed as:

$$N(r) = 2\pi n_b \int_0^r g(r') r' dr' \quad (46)$$

where r represents an arbitrary distance from a reference particle.

We present the average Radial Distribution Function ($g(r)$) and Coordination Number ($N(r)$) profiles for BF_4^- , TFSI^- , and EMIM^+ ions, as well as DMSO solvent, surrounding a single Li^+ ion. Additionally, we analyze the $g(r)$ and $N(r)$ between molecules of the same type, represented by their center-of-mass atoms. Li^+ ions preferentially coordinate with molecules containing highly electronegative atoms, i.e., atoms with a negative partial charge. The flat $g(r)$ curves in Figure 5 (e) indicate a weak interaction between Li^+ and EMIM^+ . Consequently, Li^+ primarily coordinates with BF_4^- , TFSI^- , and DMSO. According to Figure 5 (c), (d), and (f), Li^+ ions interact strongly with BF_4^- , TFSI^- , and DMSO. The most probable distances between Li^+ and the constituent atoms are: BF_4^- ($\text{Li}^+\text{-F} \approx 2.47 \text{ \AA}$, $\text{Li}^+\text{-B} \approx 3.22 \text{ \AA}$), TFSI^- ($\text{Li}^+\text{-O} \approx 2.57 \text{ \AA}$, $\text{Li}^+\text{-F} \approx 2.85 \text{ \AA}$), and DMSO ($\text{Li}^+\text{-O} \approx 2.45 \text{ \AA}$). The Li^+ cations in the electrolyte are more solvated by BF_4^- and DMSO than TFSI^- , as evidenced by the higher peaks in the $g(r)$ and the corresponding coordination number curves of $\text{Li}^+\text{-BF}_4^-$. BF_4^- and DMSO are closer to the Li^+ ions than TFSI^- ions, playing a crucial role in the dissociation of LiTFSI salt. However, TFSI^- can also access the first coordination shell of Li^+ due to strong interactions between its O and F atoms and Li^+ . The presence of both BF_4^- and TFSI^- facilitates favorable coordination between Li^+ and F, potentially leading to the formation of a Li-F solid electrolyte interphase (SEI) that protects the Li-metal anode. Interactions between molecules of the same type are weak due to electrostatic repulsion, resulting in well-dispersed molecules throughout the simulated electrolyte. The average first shell coordination number values for a tagged Li^+ cation were determined using a threshold distance of 3 $\text{\AA}$, based on the range of the first $g(r)$ peaks. Key findings from the coordination number analysis include: $\text{Li}^+\text{-Li}^+$ ($N(r = 3\text{\AA}) \approx 0$ and $N(r = 5\text{\AA}) \approx 0.25$), $\text{Li}^+\text{-BF}_4^-$ ($N_{\text{Li-F}}(r = 3\text{\AA}) \approx 2$, $N_{\text{Li-B}}(r = 3\text{\AA}) \approx 0.25$), $\text{Li}^+\text{-DMSO}$ ($N_{\text{Li-O}}(r = 3\text{\AA}) \approx 0.8$), and $\text{Li}^+\text{-TFSI}^-$ ($N_{\text{Li-O}}(r = 3\text{\AA}) \approx 0.25$ and $N_{\text{Li-F}}(r = 3\text{\AA}) \approx 0.25$).

The high ionic conductivity (Table 5) of the simulated electrolyte is derived from several key factors revealed by the solvation structure and interaction analyses (Figure 5). Li^+ cations are preferentially solvated by BF_4^- and

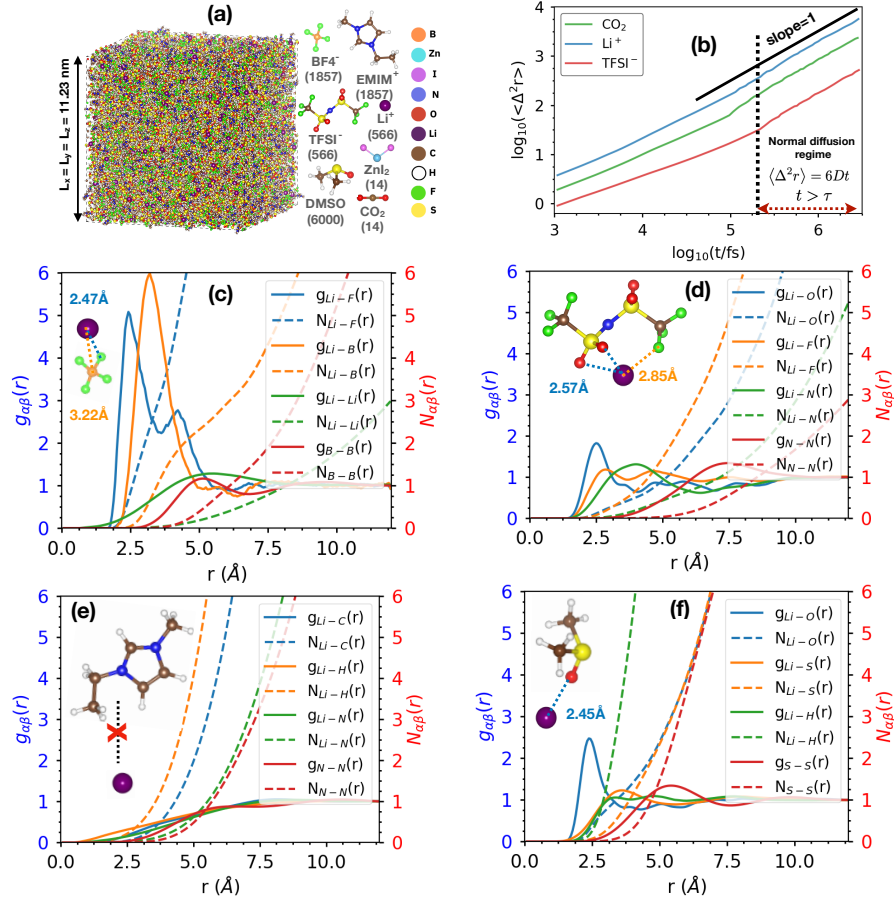


Figure 5: (a) A snapshot of the MD simulation of the liquid electrolyte, equilibrated at room temperature and ambient pressure ($T = 300$ K, $P = 1$ bar), is shown. The molecules comprising the electrolyte are depicted with their corresponding numbers in the simulation box. The atom colors used are also illustrated. (b) Mean-square-displacement of Li^+ , TFSI^- , and CO_2 within the electrolyte obtained from our MD simulation. (c), (d), (e), (f) Comparison of the average Radial Distribution Function ($g(r)$) and Coordination Number ($N(r)$) profiles for BF_4^- (c), TFSI^- (d), and EMIM $^+$ ions (e), as well as DMSO solvent (f), surrounding a single Li^+ ion. Additionally, the $g(r)$ and $N(r)$ between molecules of the same type are represented, which can be referred to as the $g(r)$ and $N(r)$ of the center-of-mass atoms of the molecules.

DMSO, promoting LiTFSI salt dissociation and increasing free charge carriers. Weak Li^+ - TFSI^- interactions facilitate easier Li^+ transport. The presence of BF_4^- and TFSI^- creates Li^+ conducting channels through favorable Li^+ -F coordination. Well-dispersed ionic species due to weak same-type molecular interactions enhance efficient ion transport. Optimal coordination numbers (e.g.,

$N_{\text{Li-F}}(r = 3\text{\AA}) \approx 2$ for BF_4^- , $N_{\text{Li-O}}(r = 3\text{\AA}) \approx 0.8$ for DMSO) allow efficient Li^+ transport while maintaining sufficient solvation. Low Li^+ - Li^+ coordination ($N(r = 3\text{\AA}) \approx 0$) minimizes clustering, promoting ion mobility. Multiple solvating species create a dynamic environment, facilitating rapid solvating molecule exchange and enhancing Li^+ mobility. The preferential solvation of Li^+ by BF_4^- and DMSO allows for more free movement compared to the larger TFSI $^-$ anions, resulting in a higher proportion of the ionic current carried by Li^+ and thus increasing the Li^+ transference number.

3.4. Cell performance from Finite Element Analysis

The simulated voltage-capacity curves of the Li- CO_2 battery cell during charge/discharge cycles are shown in Figure 6 (a). These curves were generated through Finite Element Analysis (FEA) using input parameters derived from atomistic simulations (Figure 3, Table 5) and additional material properties detailed in Tables 1, 2, and 3. The FEA results were validated against experimental data collected at a current density of 1 mAh/cm 2 , as described in the Supporting Information (SI). A strong correlation between simulated and experimental voltages is observed. During discharge, the voltage profile exhibits three distinct regions: (1) an initial sharp voltage drop due to kinetic overpotential ($\phi_1 - \phi_2 - E^0$) [70], (2) a linear decline attributed to resistive losses (ϕ_{film}) across the growing $\text{Li}_2\text{CO}_3/\text{C}$ film (average plateau potential: 2.65 V), (3) a "sudden death" characterized by rapid voltage collapse after 5000 mAh/g. The latter two phenomena, previously described for Li- O_2 batteries as charge-transport limitations through Li_2O_2 films [71], exhibit analogous behavior in Li- CO_2 systems. During charging, the cell maintains a stable potential of approximately 4.2 V until reaching 6000 mAh/g, followed by a steep voltage rise. The system achieves a specific capacity of 6200 mAh/g (0.48 mAh/cm 2).

Figure 6 (b) illustrates the impact of applied current density of 1 mA/cm 2 , 0.5 mA/cm 2 , 0.25 mA/cm 2 , and 0.1 mA/cm 2 on the simulated discharge curves of Li- CO_2 battery cell and Figure S1 shows the same curves for current densities of 0.2 mA/cm 2 , 0.3 mA/cm 2 , 0.4 mA/cm 2 , 0.6 mA/cm 2 , 0.7 mA/cm 2 , 0.8 mA/cm 2 and 0.9 mA/cm 2 . The discharge capacity exhibits a substantial reduction, decreasing from 81,570 mAh/g at a low current density of 0.1 mA/cm 2 to 6,200 mAh/g at the high current density of 1 mA/cm 2 . This observation aligns with prior experimental results, indicating that current density has a significant influence on cell capacity [72]. Furthermore, the discharge voltage plateau decreases as the current density increases (Figure 6 (b)). The capacity decrease at elevated discharge rates can be attributed to limitations in CO_2 transport through the cathode, which becomes saturated with electrolyte and unable to sustain the electrochemical reaction. Consequently, as the discharge rate increases, CO_2 reduction is confined to a localized area near the cathode-current collector interface. Furthermore, the rapid reduction in porosity due to Li_2CO_3 and C deposition on the active area's surface further impedes CO_2 transport into the cell, preventing complete utilization of the electrode's porosity.

Figure 6 (c) demonstrates that the capacity of the studied Li- CO_2 battery cell decreases exponentially with increasing current density, which can be de-

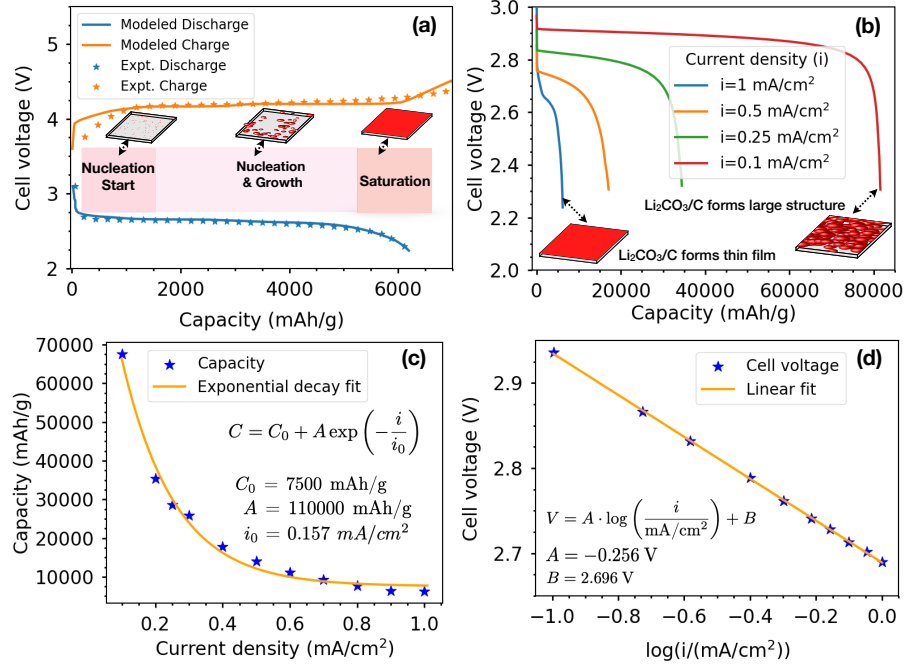


Figure 6: (a) Deep discharge/charge curve of the Li-CO₂ battery cell operating at 1 mA/cm², obtained from our multiscale model and experiment. (b) Deep discharge curves of the Li-CO₂ battery cell operating at different applied current densities of 1 mA/cm², 0.5 mA/cm², 0.25 mA/cm², and 0.1 mA/cm². The insets in (a) and (b) show a schematic representation of the Li₂CO₃/C formation process (red color) on the cathode's CO₂ feed side surface (gray color). (c) Capacity as a function of discharge current density. (d) Cell potential at the start of the discharge voltage plateau (Li₂CO₃/C nucleation and growth) with respect to current density.

scribed by the following equation [73]:

$$C = C_0 + A \exp\left(-\frac{i}{i_0}\right) \quad (47)$$

where: $C_0 = 7500$ mAh/g is the baseline capacity, $A = 110000$ mAh/g is a scaling factor, $i_0 = 0.157$ mA/cm² is a characteristic current density. At low current density ($i \ll 0.157$ mA/cm²), the capacity C tends towards its maximum value of 117500 mAh/g. This is the sum of the baseline capacity (C_0) and the additional capacity represented by A . As the current density i increases, there is a rapid decrease in the additional capacity provided by A , resulting in a decrease in the total capacity C . The parameter i_0 determines the rate at which the capacity decreases with increasing current density. A smaller i_0 would mean a faster decrease in capacity for a given increase in current density, while a larger i_0 would indicate a slower decrease. As the current density increases further, the capacity approaches the baseline value C_0 of 7500 mAh/g. This suggests

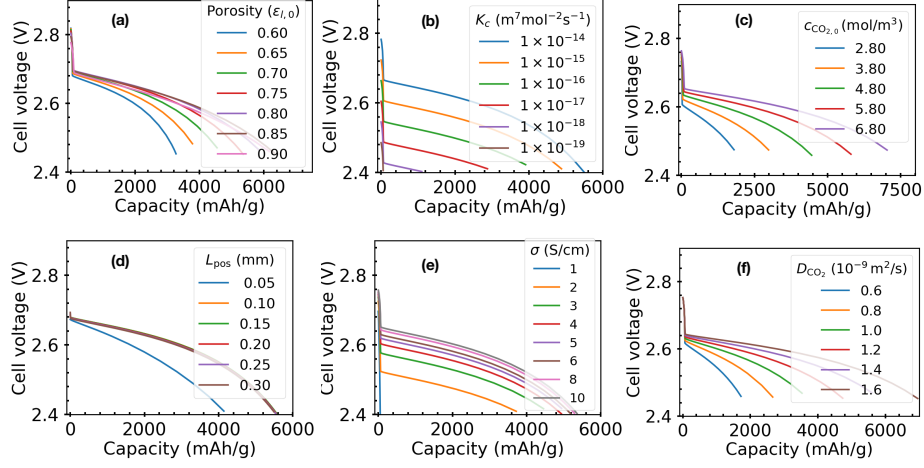


Figure 7: Impact of key parameters on voltage–capacity profiles: (a) Initial porosity ($\epsilon_{l,0}$), (b) cathodic reaction rate constant (K_c), (c) initial $c_{CO_2,0}$ concentration, (d) cathode thickness (L_{pos}), (e) ionic conductivity (σ), and (f) CO_2 diffusion coefficient (D_{CO_2}). The current density was held constant at 1 mA/cm^2 ; all other parameters are used as provided in Tables 1, 2, and 3.

that at high current densities, the battery’s capacity is primarily limited to its baseline capacity, as the contribution from the exponential term becomes negligible. The exponential decrease in capacity with increasing current density highlights a trade-off between power output and energy capacity [74]. At higher current densities ($i \gg 0.157 \text{ mA/cm}^2$), the battery can deliver more power, but at the cost of reduced capacity. This is a critical consideration in applications where both high power and high capacity are required. The connection between cell voltage (V) and current density (i) during the discharge of the studied Li- CO_2 battery cell in Figure 6 (d) is represented by the formula [73, 75]:

$$V = A \cdot \log \left(\frac{i}{\text{mA/cm}^2} \right) + B \quad (48)$$

where $A = -0.256 \text{ V}$ and $B = 2.696 \text{ V}$. The parameter A represents the slope of the voltage change with the logarithm of the current density/ (mA/cm^2) , while B represents the voltage at a reference current density of 1 mA/cm^2 . This formula indicates that as the current density increases, the cell voltage decreases logarithmically, a common characteristic in electrochemical cells due to increased polarization and decreased voltage at higher current densities [73, 75].

Figure 7 summarizes the influence of key parameters on the voltage–capacity profiles of Li- CO_2 batteries at a current density of 1 mA/cm^2 . To investigate the influence of key parameters on the voltage-capacity profiles (Figure 7), we performed a parameter sensitivity analysis using FEA to solve the model equations described in Section 2.4. The base case parameters, as provided in Tables

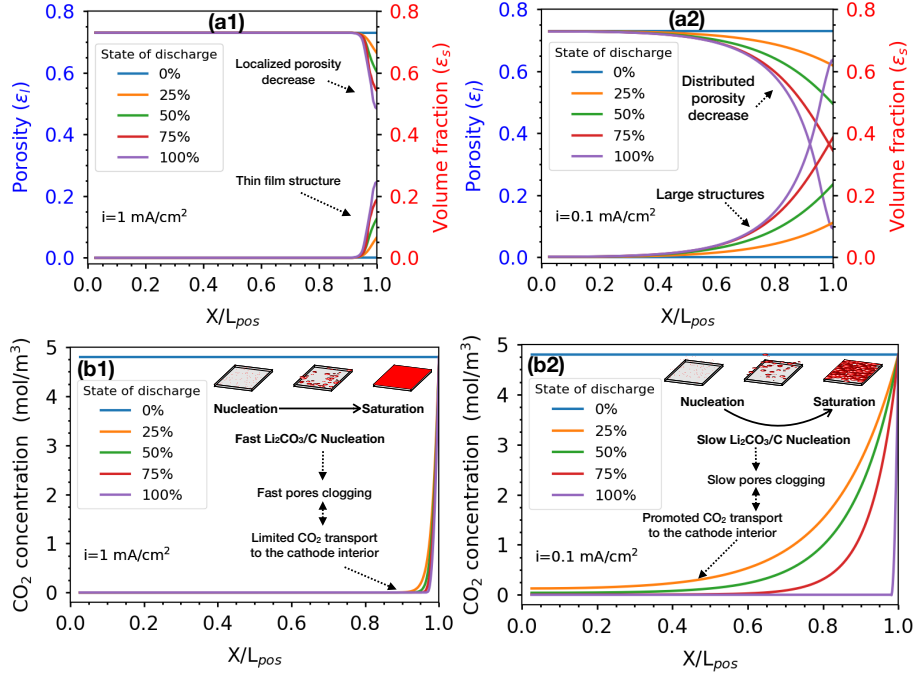


Figure 8: The changes in porosity and volume fraction of deposited species throughout the discharge process are shown at different current densities ((a1) 1 mA/cm² and (a2) 0.1 mA/cm²) in relation to the state of discharge. The local CO₂ concentration within the Li-CO₂ cathode during discharge was analyzed at rates of 0.1 mA/cm² (b1) and 1 mA/cm² (b2), corresponding to different discharge states (0% indicates a fully charged battery and 100% indicate a fully discharged battery). The distance (X) is normalized by the cathode thickness (L_{pos}). The insets in (b1) and (b2) show a schematic representation of the different nucleation processes leading to different Li₂CO₃/C formation structures depending on current density.

1, 2, and 3, were used as the reference. Each parameter was varied over a specified range while keeping the others at their base values. Specifically, the initial porosity ($\epsilon_{l,0}$) was varied from 0.6 to 0.9 in steps of 0.05, the cathodic reaction rate constant (K_c) from 1×10^{-19} to 1×10^{-14} m⁷/s/mol² (logarithmically spaced), the initial CO₂ concentration ($c_{CO_2,0}$) from 2.8 to 6.8 mol/m³, the cathode thickness (L_{pos}) from 0.05 to 0.3 mm, the ionic conductivity (σ) from 1 to 10 mS/cm, and the CO₂ diffusion coefficient (D_{CO_2}) from 0.6×10^{-9} m²/s to 1.6×10^{-9} m²/s. For each set of parameters, the discharge process was simulated at a constant current density of 1 mA/cm². As shown in Figure 7 (a), increasing the initial porosity ($\epsilon_{l,0}$) leads to higher battery capacity by expanding the available surface area and enhancing CO₂ accessibility, thereby promoting more efficient electrochemical reactions. Figure 7 (b) highlights the critical role of the cathodic reaction rate constant (K_c); the voltage–capacity profile exhibits high sensitivity to variations in K_c , emphasizing the necessity of

effective catalytic materials for optimal battery performance. In Figure 7 (c), it is evident that higher initial CO_2 concentration ($c_{\text{CO}_2,0}$) results in increased capacity and a more stable voltage plateau, attributed to improved reactant availability at the cathode. Figure 7 (d) demonstrates that increasing the cathode thickness (L_{pos}) from 0.05 mm to 0.1 mm substantially enhances capacity due to the greater active volume; however, further increases beyond 0.1 mm do not provide additional capacity gains, likely due to diffusion limitations. Figure 7 (e) shows that electrolyte ionic conductivity (σ) is a critical factor, with values above 2 mS/cm required to achieve high capacities (>4000 mAh/g) and improved voltage stability. Figure 7 (f) indicates that a higher CO_2 diffusion coefficient (D_{CO_2}) further enhances both capacity and voltage plateau by facilitating more efficient CO_2 transport to active sites. Notably, the same trends are observed at lower current densities; Figure S2 depicts these effects at a current density of 0.4 mA/cm². Taken together, these results underscore the importance of simultaneously optimizing electrode structure, electrolyte properties, and catalytic activity to maximize the performance of Li- CO_2 batteries.

Figure 8 (a1) and (a2) illustrate how porosity (ε_l) and volume fraction of $\text{Li}_2\text{CO}_3/\text{C}$ deposited species (ε_s) change over space and time with current densities of $i=1$ mA/cm² and $i=0.1$ mA/cm², respectively. Figure S3 presents a comparison of these parameters at $i=0.1$, $i=0.25$, $i=0.5$, and $i=1$ mA/cm². For high current density ($i=1$ mA/cm²), porosity decreases gradually, primarily near the CO_2 feed side of the cathode ($X/L_{\text{pos}}=1$), from its initial value of $\varepsilon_l = 0.73$ to 0.48 at the end of discharge. This decrease is due to the formation of a thin film (flake-like) of $\text{Li}_2\text{CO}_3/\text{C}$ [76], represented by the gradual increase of ε_s from 0 to 0.25 near the cathode feed side. This film formation leads to electrode clogging, impeding efficient CO_2 diffusion into the cathode. For low current density ($i=0.1$ mA/cm²), $\text{Li}_2\text{CO}_3/\text{C}$ forms larger structures [76], evidenced by the gradual increase of ε_s across a wider region of the cathode from 0 to 0.64. This results in a distributed porosity decrease from 0.73 to 0.09, initially allowing more CO_2 transport into the cathode. However, as discharge progresses, the substantial porosity decrease hinders CO_2 diffusion even at low current densities.

Figure 8 (b1) and (b2) present the CO_2 concentration profiles within the cell during discharge at $i=1$ mA/cm² and $i=0.1$ mA/cm², respectively. Figure S4 compares concentration profiles at various current densities. At $i=1$ mA/cm², the CO_2 concentration (c_{CO_2}) decreases sharply from its initial value of 4.8 mol/m³ to 0 mol/m³ in the porous cathode, indicating strongly impeded CO_2 diffusion. At $i=0.1$ mA/cm², the decrease is more gradual. The impediment to CO_2 transport, which is inversely proportional to the current density, results from the accumulation of Li_2CO_3 and C on the active cathode surface. This accumulation reduces pore availability for the electrolyte, increases the amount of insulating species, and obstructs pores, collectively restricting CO_2 transport and limiting cell capacity (Figure 6 (c)). The nucleation and growth of Li_2CO_3 significantly impact performance, as described in Figure 4 and the detailed FEA model in section 2.4. Current density plays a crucial role in this process. At high current densities (1 mA/cm²), rapid nucleation near the CO_2 feed side forms a thin $\text{Li}_2\text{CO}_3/\text{C}$ film (Figure 8 (a1), (b1)). In contrast, low current densities (0.1

mA/cm²) promote slower growth, resulting in larger, more distributed Li₂CO₃ structures and a gradual decrease in porosity (Figure 8 (a2), (b2)).

To address these limitations, an ideal cathode should feature: a large surface area for reaction sites and product deposition, large pores to facilitate active species transport and prevent clogging, high porosity to accommodate discharge product growth, and high electrical conductivity and stability. Such a cathode design could mitigate the observed limitations and potentially enhance the overall performance of Li-CO₂ batteries [77]. Finally, we note that direct experimental validation is essential to confirm our model’s mechanistic predictions. To rigorously assess accuracy and applicability, high-resolution imaging techniques such as transmission electron microscopy (TEM) and scanning electron microscopy (SEM) should be used to verify predicted changes in Li₂CO₃/C morphology, cathode surface features, and Li₂CO₃/C aggregation states. Additionally, time-resolved or ex-situ X-ray diffraction (XRD) can track the evolution of Li₂CO₃/C crystal structure and phase composition, while gas adsorption measurements, including Brunauer-Emmett-Teller (BET) and Barrett-Joyner-Halenda (BJH) analysis, can provide a quantitative assessment of cathode porosity and specific surface area. Importantly, a multiscale experimental approach is also needed to capture material behavior across different length and time scales, ensuring that both nanoscale mechanisms and macroscopic properties are accurately linked and described. These targeted and multiscale experiments are crucial for establishing the model’s predictive accuracy and real-world relevance.

4. Conclusion

Our interactive multiscale modeling framework, integrating FEA, MD, and DFT/AIMD, provides comprehensive insights into Li-CO₂ battery electrochemistry. This approach enables accurate prediction of cell performance by considering critical factors such as CO₂ transport, cathode porosity, and Li₂CO₃/C deposition. DFT/AIMD calculations reveal the high electrical conductivity of the cathode catalyst, while MD simulations demonstrate favorable electrolyte properties. FEA results show an exponential decrease in discharge capacity with increasing current density—from 81,570 mAh/g at 0.1 mA/cm² to 6,200 mAh/g at 1 mA/cm²—primarily due to CO₂ transport limitations and reduced cathode porosity. The morphology of Li₂CO₃/C deposits transitions from larger structures at lower current densities to thin films at higher current densities, significantly impacting performance.

Our multiscale approach bridges atomic-level phenomena and cell-level performance, offering unique insights into the spatiotemporal evolution of key battery parameters and reducing the need for extensive experimental prototyping. Future research should focus on enhancing CO₂ transport, improving discharge product decomposition, and optimizing cathode design with large surface areas, optimal pore sizes, and high porosity. This framework accelerates the development of high-performance Li-CO₂ batteries and sustainable energy storage technologies by providing insights into phenomena that are difficult to observe

experimentally in real time. Moreover, the developed Li-CO₂ battery continuum model at the cell level can be extended to study battery packs and thermal runaway. This modeling framework can also be adapted to other battery chemistries or used to simulate degradation, cycling behavior, and thermal effects, thereby broadening its applicability.

CRedit Authorship Contribution Statement

Mohammed Lemaalem, Selva C. Selvaraj, and Naveen K. Dandu: Conceptualization, Visualization, Software, Data Curation, Formal Analysis, Methodology, Writing – Original Draft, Writing – Review & Editing. Ilias Papailias and Arash Namaeighasemi: Methodology, Data Curation, Formal Analysis, Writing – Review & Editing. Larry A. Curtiss and Amin Salehi-Khojin: Supervision, Data Curation, Formal Analysis, Writing – Review & Editing. Anh T. Ngo: Supervision, Conceptualization, Data Curation, Formal Analysis, Investigation, Project Administration, Writing – Review & Editing.

Author Contributions

This work was conceived by A. T. N., M. L., S. C. S., and N. K. D., carried out the calculations, including first-principles DFT, AIMD results, large-scale MD simulations, and Finite Element Analysis. I. P., A. N., L. A. C., and A. S. K. were responsible for the experimental measurements. All authors contributed to the discussion of the results and the final manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Background on catalyst/electrolyte developments for Li-CO₂ battery; Description of the experimental setup; Computational details; Plots showing the voltage-capacity curve during Li-CO₂ battery discharge at various rates ranging from 0.2 mA/cm² to 0.9 mA/cm²; Plots showing the effect of initial values of relevant variables on the voltage-capacity curve during Li-CO₂ battery discharge at a rate of 0.4 mA/cm²; Plots illustrating changes in porosity and volume fraction of deposited species in the cathode throughout the discharge process at different current densities and discharge states; and Plots showing the change in CO₂ concentration within the cathode.

Data availability

The data that support the plots within this paper and other findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Interactive Multiscale Modeling to Bridge Atomic Properties and Electrochemical Performance in Li-CO₂ Battery Design

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1 Background: Catalyst/electrolyte developments for Li-CO₂ battery.

The Li-CO₂ battery chemistry faces critical challenges in catalyst performance, primarily characterized by two fundamental issues, catalyst inefficiency and redox mediator limitations. Noble and transition metal catalysts (such as Pt and Au nanoparticles) demonstrate low catalytic activities for CO₂ reduction/evolution reactions in aprotic media, while 2D materials (e.g., MoS₂) exhibit poor structural stability at high current rates [26]. Existing catalysts struggle to maintain consistent performance across multiple charge-discharge cycles, often due to catalyst deactivation caused by self-degradation or environmental factors such as discharge product accumulation and species contamination [26]. Unlike Li-O₂

batteries, Li-CO₂ battery liquid catalysts are typically inactive in decomposing discharge products (Li₂CO₃ and solid carbon), and current redox mediators fail to effectively facilitate the reversible transformation of reaction intermediates [5]. The formation of Li₂CO₃, a wide-bandgap insulator, leads to slow kinetics and high voltage platforms (> 4.3 V) during the charging process, significantly decreasing the round-trip efficiency of the battery [14]. The lack of efficient decomposition mechanisms significantly impedes battery cycling efficiency, with most current Li-CO₂ batteries only capable of operating for dozens of cycles with low overpotential, even with state-of-the-art catalysts [26]. These challenges necessitate developing novel catalytic systems that can maintain high catalytic activity across extended cycling, demonstrate structural stability at elevated current rates, enable efficient decomposition of discharge products, and provide reversible electrochemical reactions with minimal energy loss. The development of catalysts with high catalytic activity and conductivity is crucial to the research of high-performance Li-CO₂ batteries [12]. Promising research directions in Li-CO₂ battery technology encompass a range of innovative approaches aimed at overcoming current limitations and enhancing overall performance [23, 14, 19]. Advanced solid redox mediators have shown significant potential, as demonstrated by a study using a Cu(II) coordination compound of benzene-1,3,5-tricarboxylic acid. This solid redox mediator enabled a Li-CO₂ battery to achieve a higher discharge voltage of 2.8 V, a lower charge potential of 3.7 V, and superior cycling performance over 400 cycles [23]. The use of solid redox mediators addresses the shuttle effect and sluggish kinetics associated with soluble redox mediators, potentially improving long-term stability and efficiency. Alternative reaction pathway engineering has emerged as another promising direction, focusing on promoting the formation of Li₂C₂O₄ instead of Li₂CO₃ as the discharge product. This approach can reduce the charge potential and improve overall battery performance [14]. For instance, the use of Mo₂C as a cathode material has been shown to stabilize Li₂C₂O₄ as the final discharge product, reducing the charge potential to 3.8 V and enhancing the reversibility of the battery [14]. Innovative catalyst designs, including carbon-based and metal-complex catalysts, are being extensively

explored. Carbon-based materials, such as multi-walled carbon nanotubes (MWCNTs), are widely used as cathodes due to their high electrical conductivity and large surface area [19]. However, their limited catalytic activity for Li_2CO_3 decomposition has led to investigations of ways to enhance their performance, such as introducing metal catalysts like Ru to reduce overpotentials and improve efficiency [19]. Transition metal compound catalysts have shown promise in improving the catalytic activity and stability of Li- CO_2 batteries. For example, the integration of Ru-catalyzed MWCNT cathodes with solid electrolytes has demonstrated enhanced electrochemical reactions involving CO_2 [19]. Additionally, nanostructured catalyst development with improved conductivity and catalytic properties continues to be a focus area, exploring various materials and structures to optimize battery performance [9, 25, 18].

Ionic liquid-based electrolytes show promise for Li- CO_2 batteries due to their stability with intermediate products compared to solid-state electrolytes, but their development is limited by low ionic conductivity and high viscosity [26]. To address these limitations, ionic liquid electrolytes based on DMSO (dimethyl sulfoxide) as a solvent and LiTFSI (lithium bis(trifluoromethanesulfonyl)imide) as a salt, often incorporating EMIM- BF_4 (1-Ethyl-3-methylimidazolium tetrafluoroborate), have been developed [15, 27, 21, 24, 1, 10, 9]. These electrolytes are preferred for Li- CO_2 batteries due to their superior ionic conductivity, enhanced electrochemical stability, and improved CO_2 activation properties. DMSO-based electrolytes exhibit faster Li^+ ion diffusion kinetics compared to other solvents like TEGDME (tetraethylene glycol dimethyl ether), resulting in lower polarization potential [15]. Li- CO_2 batteries using 3M LiTFSI in DMSO electrolyte have demonstrated excellent cycling performance, maintaining stable operation for over 1,900 hours (180 cycles) [15]. In concentrated DMSO-based electrolytes, TFSI $^-$ anions can access the primary solvation sheath of Li^+ ions, leading to the formation of a LiF-rich solid electrolyte interphase (SEI) layer, which is beneficial for battery performance [27, 21]. DMSO-based electrolytes demonstrate good electrochemical stability with both the Li metal anode and the reactive cathode at high volt-

ages and elevated temperatures [15]. The use of DMSO-based electrolytes improves the rate capability of the battery, with cells able to deliver flat charge/discharge plateaus at current densities of up to 2,000 mA g⁻¹ [27]. Together, these properties make the DMSO LiTFSI electrolyte a promising choice for improving the overall performance and stability of Li-CO₂ batteries. The addition of EMIM-BF₄ (1-Ethyl-3-methylimidazolium tetrafluoroborate) as an electrolyte material plays several important roles in Li-CO₂ batteries. EMIM-BF₄ helps in activating CO₂ molecules, which is crucial for the CO₂ reduction reaction in Li-CO₂ batteries [24]. The addition of EMIM-BF₄ to the electrolyte has been shown to enhance battery performance. In one study, a Li-CO₂ battery using 0.1 M LiTFSI in (EMIM-BF₄)/DMSO as the electrolyte achieved 500 cycles at 0.5 A/g with a capacity of 500 mAh/g [15]. The presence of EMIM-BF₄ in the electrolyte contributes to improved cycling stability of Li-CO₂ batteries [1]. When combined with other electrolyte components like DMSO and LiTFSI, EMIM-BF₄ contributes to the overall performance enhancement of Li-CO₂ batteries, including higher discharge voltages and increased capacities [10]. Recently, Jaradat et al. used an electrolyte combination of ZnI₂, LiTFSI, DMSO, and EMIM-BF₄ with a volumetric ratio of 2:3 that led to good Li-CO₂ battery performance [9].

2 Supplementary notes: Experimental setup

2.1 Cathode preparation

To synthesize Sb_{0.67}Bi_{1.33}Te₃ nanoflakes (NFs), a powder sample weighing 100 mg was dispersed in 20 ml of isopropyl alcohol (IPA) and subjected to ultrasonication for a duration of 24 hours. Subsequently, the solution was centrifuged at 1000 RPM for 15 minutes, allowing for the extraction of the supernatant materials. The resulting Sb_{0.67}Bi_{1.33}Te₃ was then deposited onto carbon cloth with a mass loading of 0.4 mg and subsequently dried under ultraviolet (UV) light.

2.2 Electrolyte preparation

A mixture of 1M LiTFSI and 0.025M ZnI_2 was combined with a solution of DMSO and EMIM-BF₄ ionic liquid in a ratio of 3:2. The preparation of the electrolyte was conducted within an argon-filled glove box, maintaining oxygen and humidity levels below 0.1 ppm to prevent contamination.

2.3 Battery assembly

The assembly of the battery was conducted within an argon-filled glove box, maintaining oxygen and humidity levels below 0.1 ppm. The process for assembling the 2032-type coin-cell battery was as follows: (i) the wave spring and spacer were inserted into the coin cell; (ii) a glass-fiber separator, along with 50 μl of electrolyte, was affixed to the anode; (iii) cathodes coated with $\text{Sb}_{0.67}\text{Bi}_{1.33}\text{Te}_3$ nanofibers were attached to the separator. Subsequently, holes were drilled, and the case assembly was secured to the top of the cathode. For the battery studies, a Galvanostatic Battery Analyzer (MTI Corp, BST8-MA) was employed. The cycling experiments were conducted within small, sealed bags filled with CO_2 at an ambient temperature of 25 °C.

2.4 Three electrodes set up

A three-electrode configuration was employed for conducting electrochemical Linear Sweep Voltammetry (LSV) experiments. The working electrode consisted of $\text{Sb}_{0.67}\text{Bi}_{1.33}\text{Te}_3$ nanofibers (NFs) coated cathodes, while lithium metal disks served as both the counter and reference electrodes. The electrolyte utilized for the electrochemical measurements was saturated with carbon dioxide (CO_2). The experiments were performed using a Voltalab potentiostat at a sweep rate of 5 mV/s.

3 Supplementary notes: Simulation detail

3.1 Detailed DFT Computational Parameters

DFT calculations employed the projector augmented wave (PAW) method to describe the valence electrons of Bi, Sb, Te, Li, C, and O atoms, using a plane-wave basis set with a kinetic energy cutoff of 500 eV [2]. The generalized gradient approximation of Perdew and Wang was used for exchange-correlation energy [20, 3]. The unit cells of both $\text{Sb}_{0.67}\text{Bi}_{1.33}\text{Te}_3$ and Li_2CO_3 were expanded to lattice constants greater than 20 Å to allow sampling of $1 \times 1 \times 1$ k-meshes. Structural optimization was performed by alternately minimizing ionic and electronic energies until all forces were less than ± 10 meV/Å. For surface calculations, a k-mesh of $4 \times 4 \times 1$ was used, with a total energy convergence criterion of 1×10^{-5} eV and force convergence below 0.001 eV/Å. The D3 van der Waals correction by Grimme was included in all calculations to accurately capture dispersion interactions, which are particularly important for modeling the physisorption of CO_2 on the catalyst surface.

3.2 Detailed MD Simulation Parameters

MD simulations employed a 14 Å cutoff for both Lennard-Jones and Coulomb interactions and a 1 fs time step. Long-range electrostatics were treated using the PPPM solver with an accuracy of 10^{-4} [17, 8]. Molecular geometries were optimized using the General Amber Force Field (GAFF) and Universal Force Field (UFF), and final GROMOS parameters for TFSI^- , DMSO, EMIM^+ , BF_4^- , and CO_2 were obtained from the Automated Topology Builder (ATB) [16, 22, 13]. The ATB parameter generation considers Lennard-Jones parameters refined against experimental solvation and pure liquid properties, parametrization and validation using a single-range 14 Å cutoff, atomic charges fitted to quantum mechanical electrostatic potentials, and bonded parameters assigned using force constants estimated at the B3LYP/6-31G* level of theory, resulting in robust symmetry routines for assigning identical parameters to chemically equivalent atoms, bonds, and angles. UFF was used for

ZnI_2 , and Li^+ parameters were taken from the CL&P force field [4, 6]. Initial configurations were generated with Moltemplate [11] and energy minimized [7]. The system was agitated using the Langevin thermostat at 900 K for 1 ns, then with the Nosé-Hoover thermostat. Equilibration was performed in stages: cooling from 900 K to 300 K at 500 bar for 3 ns; reducing pressure from 500 bar to 1 bar at 300 K for 3 ns; and finally, equilibrating at 300 K and 1 bar for 10 ns.

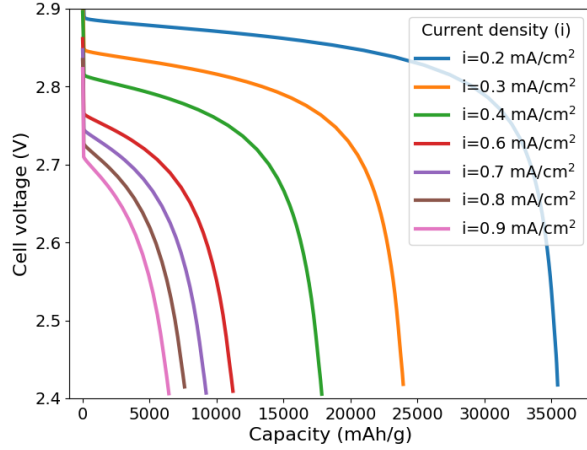
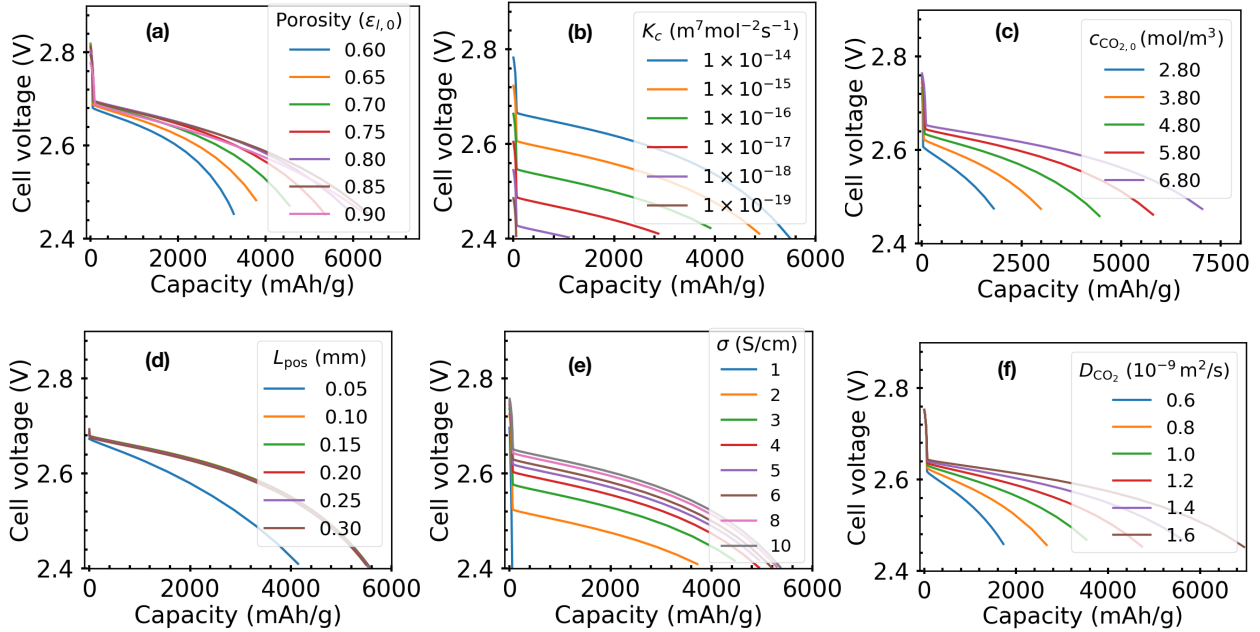


Figure S1: Voltage-capacity curve during Li-CO₂ battery discharge at various rates ranging from 0.2 mA/cm² to 0.9 mA/cm² supporting the data presented in Figure 3 (b), (c), and (d).



Impact of key parameters on voltage-capacity profiles: (a) Initial porosity ($\epsilon_{l,0}$), (b) cathodic reaction rate constant (K_c), (c) initial $c_{CO_2,0}$ concentration, (d) cathode length (L_{pos}), (e) ionic conductivity (σ), and (f) CO₂ diffusion coefficient (D_{CO_2}). The current density was held constant at 0.4 mA/cm²; all other parameters are used as provided in Tables 1, 2, and 3. We note that the same qualitative behavior is observed for the other current densities considered in this work.

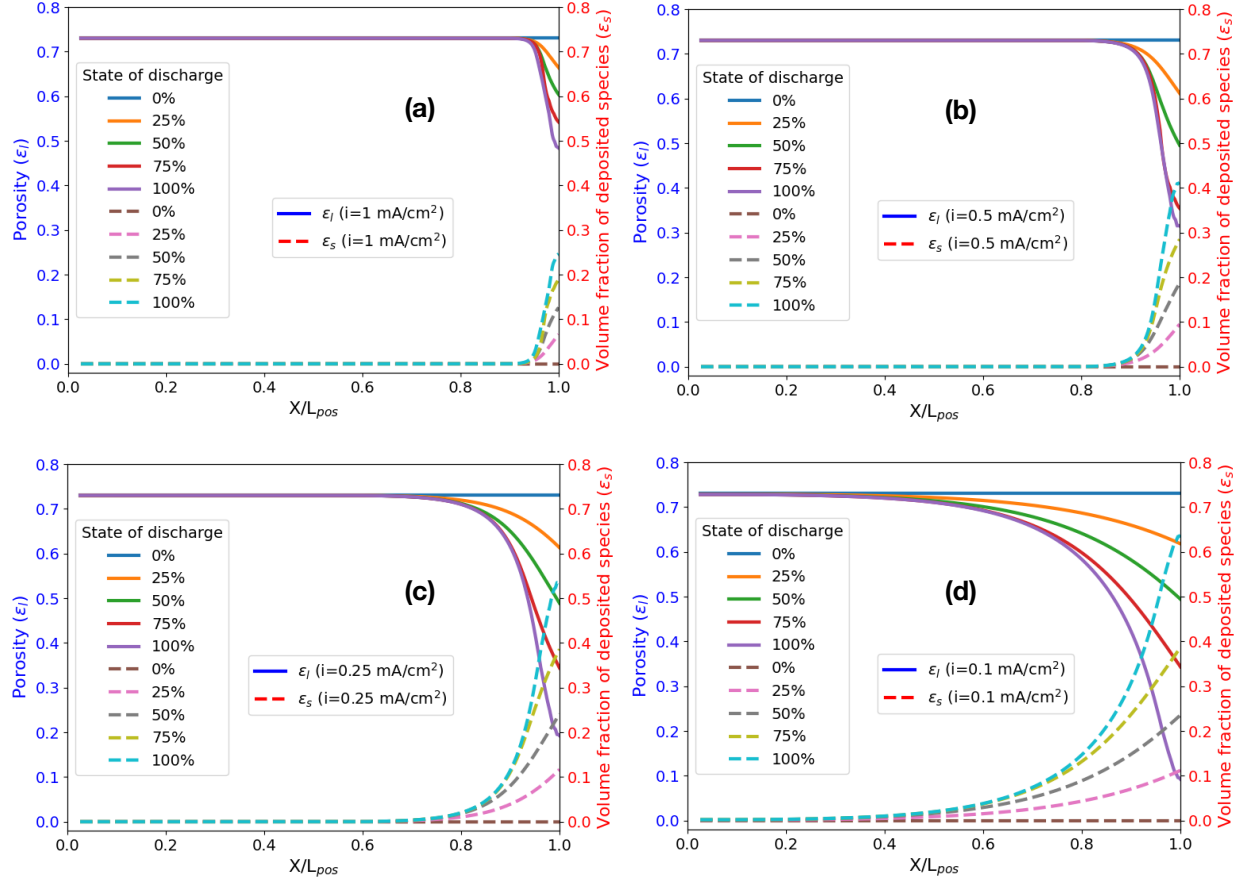


Figure S3: The change in porosity and volume fraction of deposited species in the cathode throughout the discharge process at different current densities ((a) 1, (b) 0.5, (c) 0.25, and (d) 0.1 mA/cm²) in relation to the state of discharge. The distance (x) is normalized by the cathode length (L_{pos}).

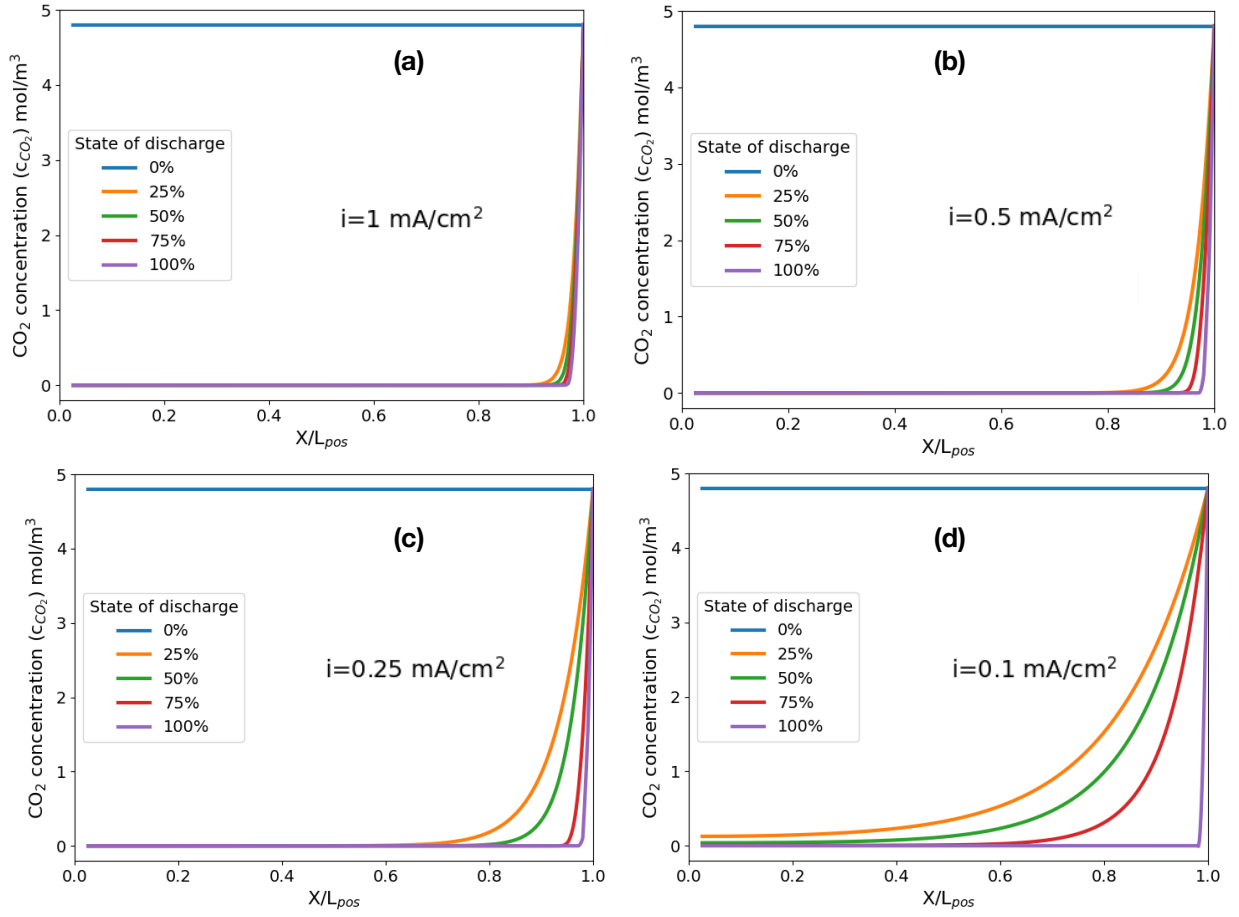


Figure S4: The local CO_2 concentration within the Li- CO_2 cathode during discharge was analyzed at various rates of 0.1, 0.25, 0.5, and 1 mA/cm^2 , corresponding to different discharge levels (0 % indicates a fully charged battery). The distance (X) is normalized by the cathode length (L_{pos}).

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Graphical abstract

